

Lab Requirements & Base System Configuration

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1. Objective

The objective of this lab is to install and configure Android Studio IDE and the Android Emulator on different Operating Systems- Linux and Windows.

2. Prerequisites

Operating System	Ubuntu 20.04 / Windows 10
Hardware	● CPU: x86_64 CPU architecture ● RAM: 8 GB RAM or more ● Disk Space: 8 GB of available disk space minimum (IDE + Android SDK + Android Emulator)
Tools Required	Not Required

3. Problem Statement

To set up the environment needed to develop the Android Applications using the Integrated Development Environment named Android Studio and to test the developed applications using the Android Virtual Device (AVD) on different Operating systems (Linux / Windows) and get familiar with the tools.

4. Summary

By implementing the steps from this module we will be able to successfully configure and install android studio in linux and windows systems.

5. Template for each step

Step 1: Installing and configuring on Linux System

1. Update the system

1.1 Open the Terminal and execute the following command.

\$ sudo apt-get update

```
ubuntu@ubuntu2004: ~
ubuntu@ubuntu2004:~$ sudo apt-get update
Hit:1 http://us.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [909 kB]
Get:6 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [1,258 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [291 kB]
```

2. Install the Java Development Kit

2.1 Open the Terminal and execute the following command to install the JDK.

\$ sudo apt install openjdk-17-jdk

```
cdac@cdac-ThinkStation-P330:~$ sudo apt install openjdk-17-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 liblvm2 libsnbios-c2 windsurf
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsn-de
  openjdk-17-jdk-headless openjdk-17-jre openjdk-17-jre-headless xiiprotoc-dev xorg-sgml-doctools xtrans-dev
Suggested packages:
  default-jre libice-doc libx11-doc libxcb-doc libxt-doc openjdk-17-demo openjdk-17-source visualvm fonts-ipafont-gothic fonts-ip
The following NEW packages will be installed:
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsn-de
  openjdk-17-jdk openjdk-17-jdk-headless openjdk-17-jre openjdk-17-jre-headless xiiprotoc-dev xorg-sgml-doctools xtrans-dev
0 upgraded, 20 newly installed, 0 to remove and 44 not upgraded.
Need to get 126 MB of archives.
After this operation, 200 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 java-common all 0.72build2 [6,782 B]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 openjdk-17-jre-headless amd64 17.0.20+8-1-22.04 [48.4 MB]
1% [2 openjdk-17-jre-headless 6,944 B/48.4 MB 0%]
```

2.2 When asked to continue, press **Y** and wait till the progress is completed.

2.3 Verify the java installation by executing the following command. The command should print the installed java version on the screen.

\$ java --version

```
cdac@cdac-ThinkStation-P330:~$ java --version
openjdk 17.0.20 2026-07-21
OpenJDK Runtime Environment (build 17.0.20+8-1-22.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.20+8-1-22.04-Ubuntu, mixed mode, sharing)
cdac@cdac-ThinkStation-P330:~$
```

3. Install the Android Studio IDE

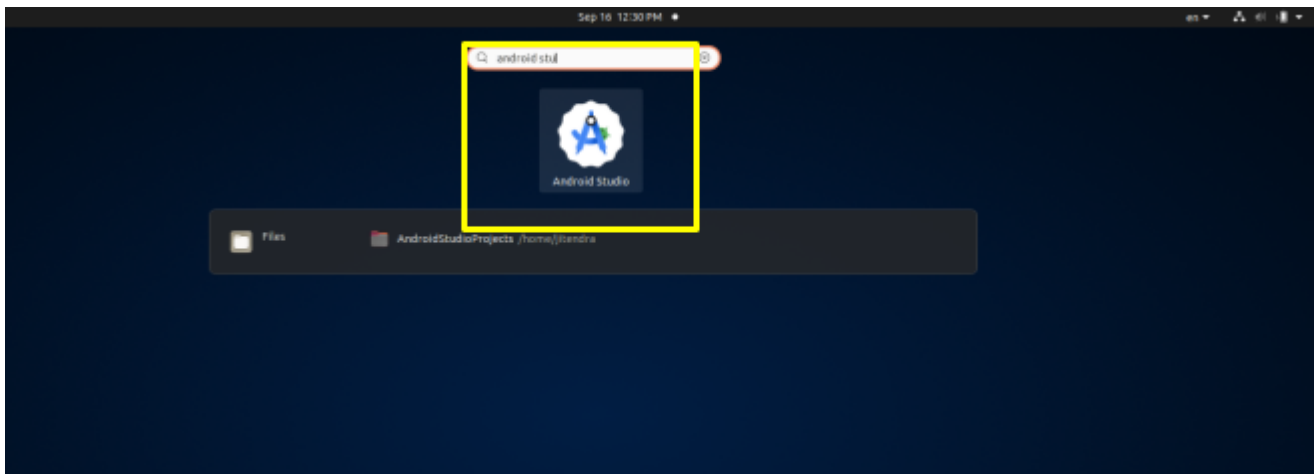
3.1 Execute the following command in the terminal to install Android Studio IDE.

\$ sudo snap install android-studio --classic

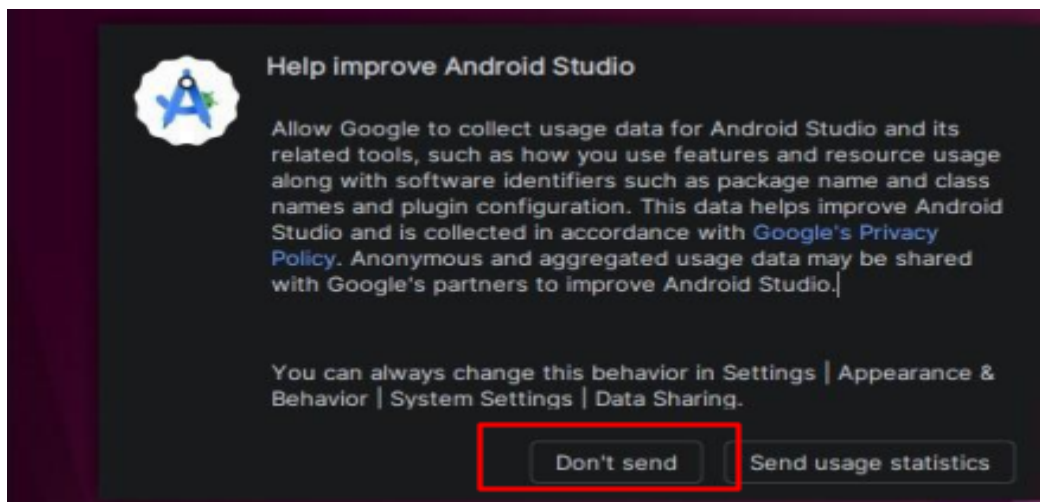
```
cdac@cdac-ThinkStation-P330:~$ sudo snap install android-studio --classic
Download snap "android-studio" (242) from channel "stable"
10% 6.88MB/s 3m3s
```

4. Configure the Android Studio IDE

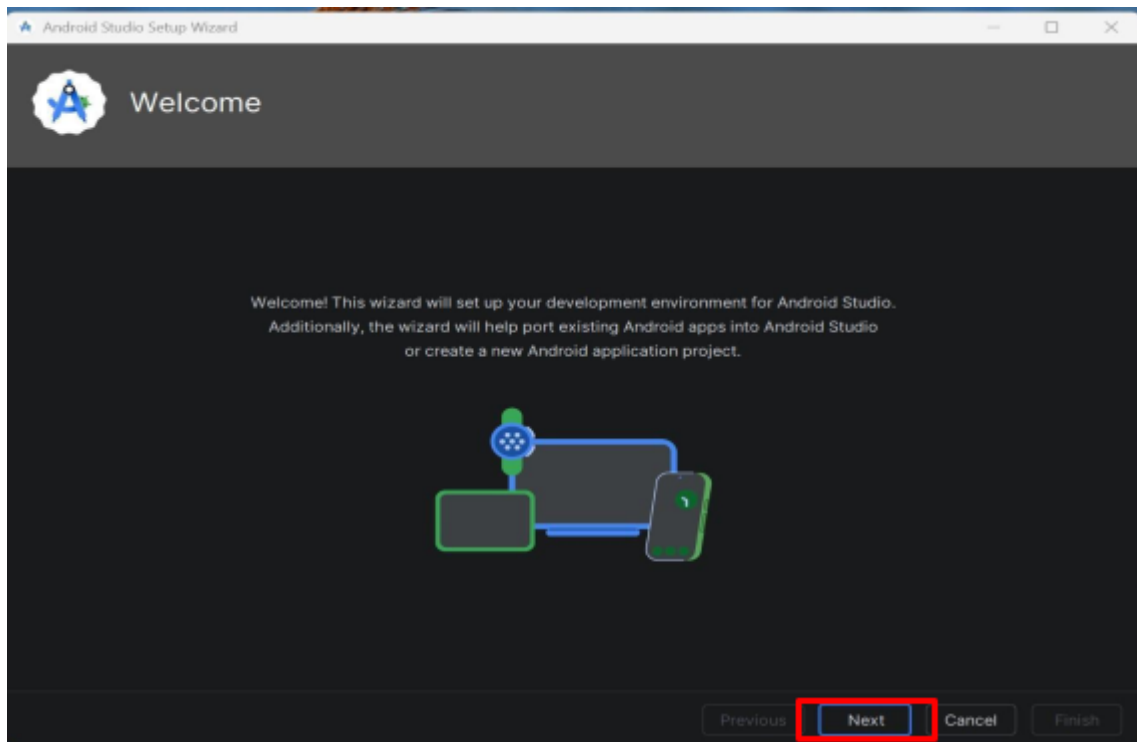
4.1 Once the installation of Android Studio is complete, start the application from the **Activities** menu by searching for the android keyword.



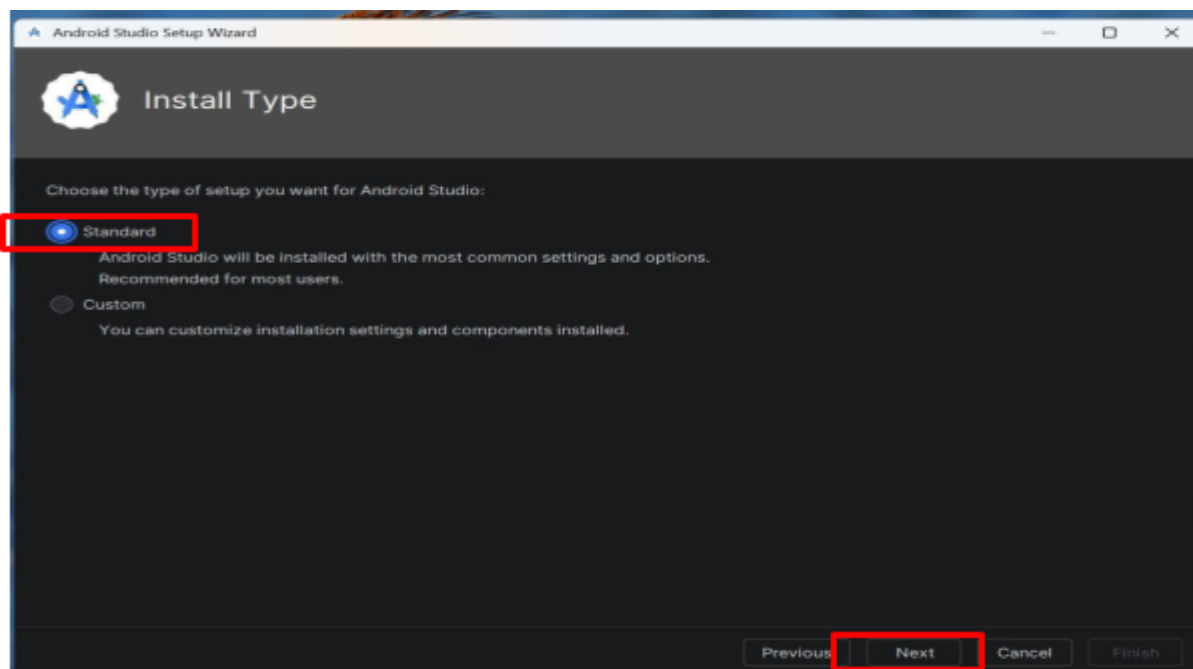
4.2 Choose between whether to **send usage statistics to Google** or not.



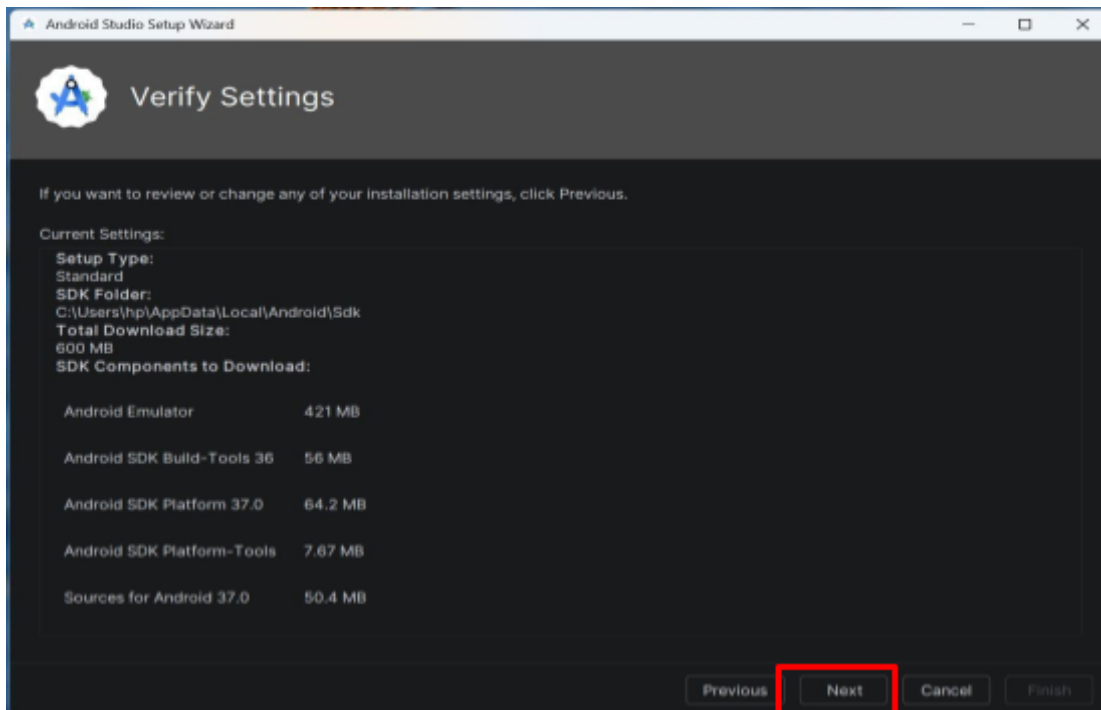
4.3 Now the Setup Wizard will open like the following. Click on the **Next** button.



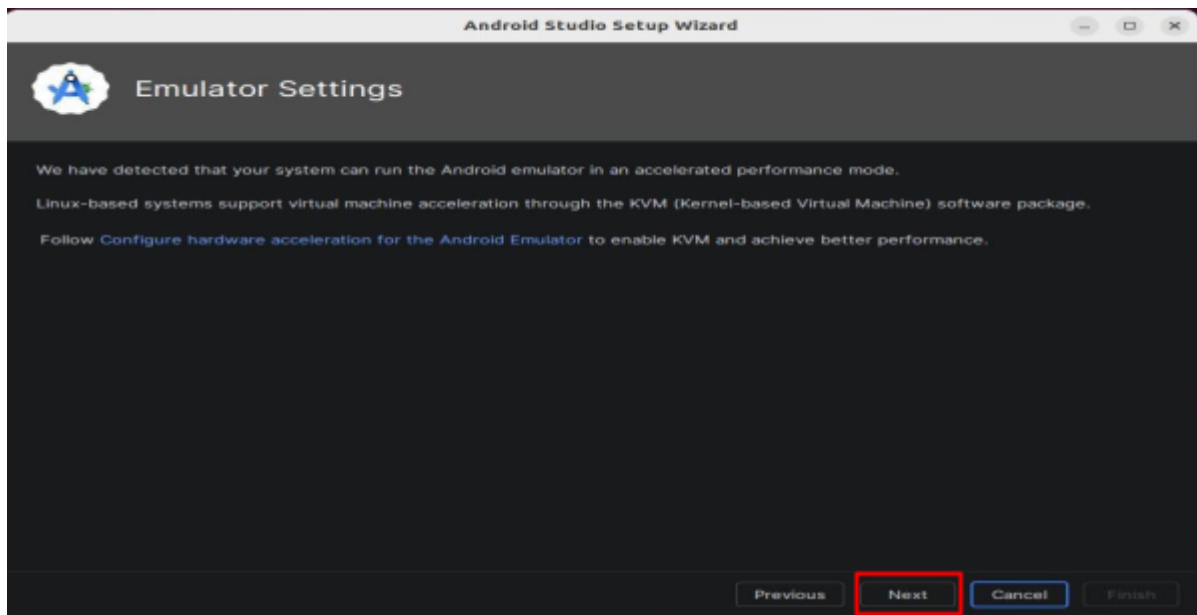
4.4 Choose Installation Type as **Standard** and then click on **Next** button.



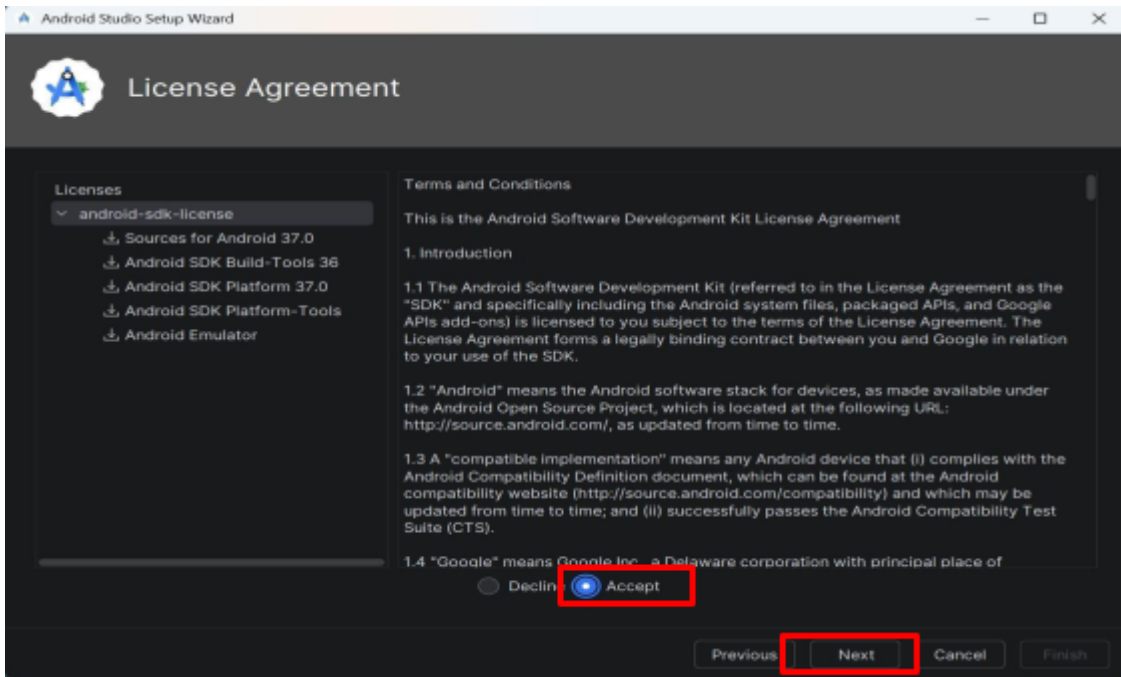
4.5 Now, click on the **Next** button to download all the SDK components.



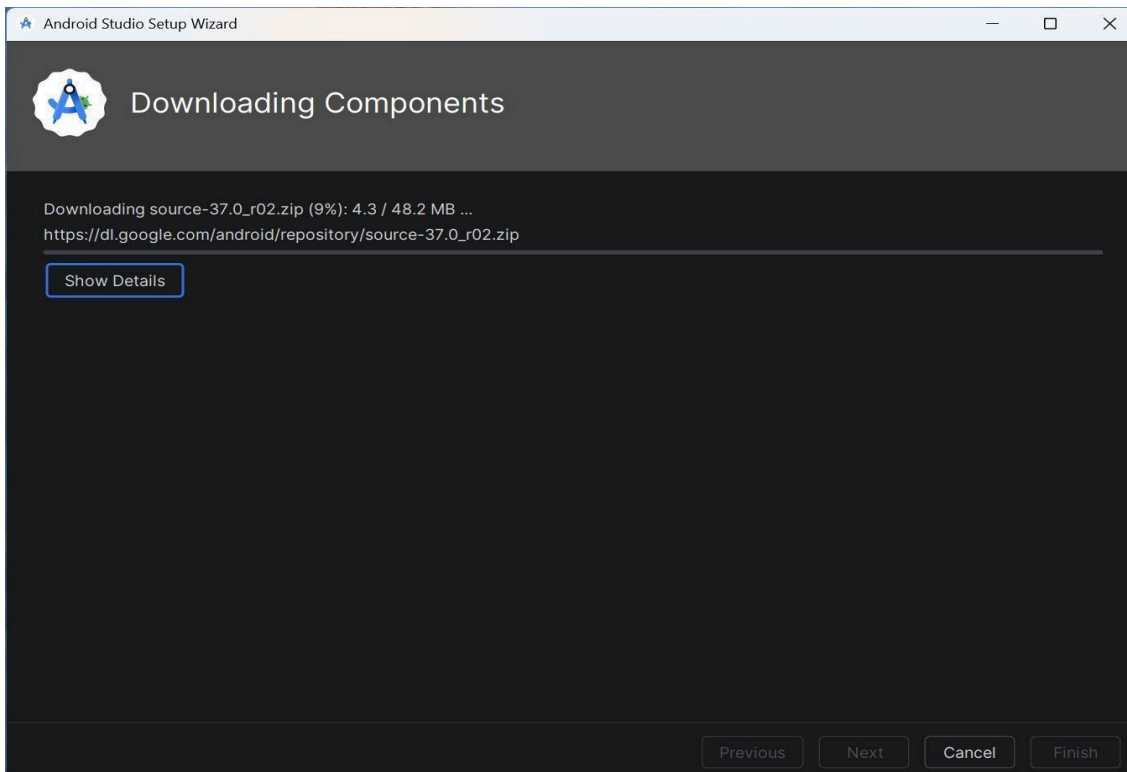
4.6 Now an Emulator Settings pop up will be displayed click on **Next**



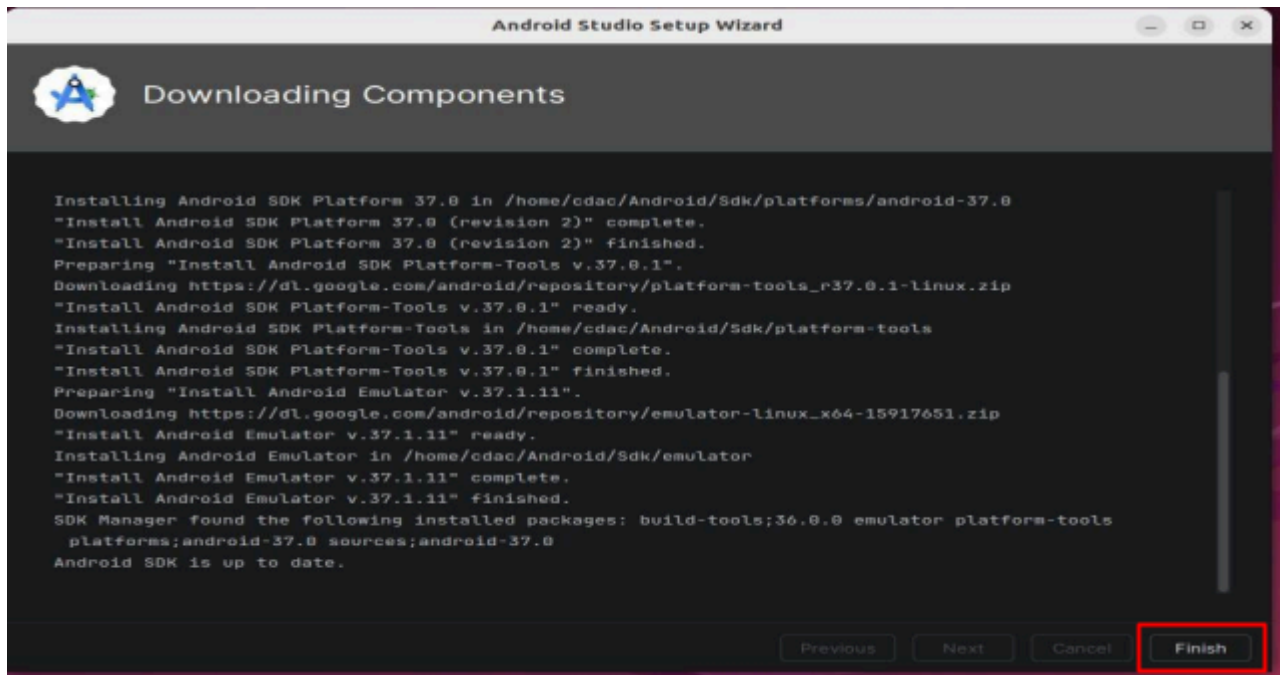
4.7 Select **Accept** and click on **Next**



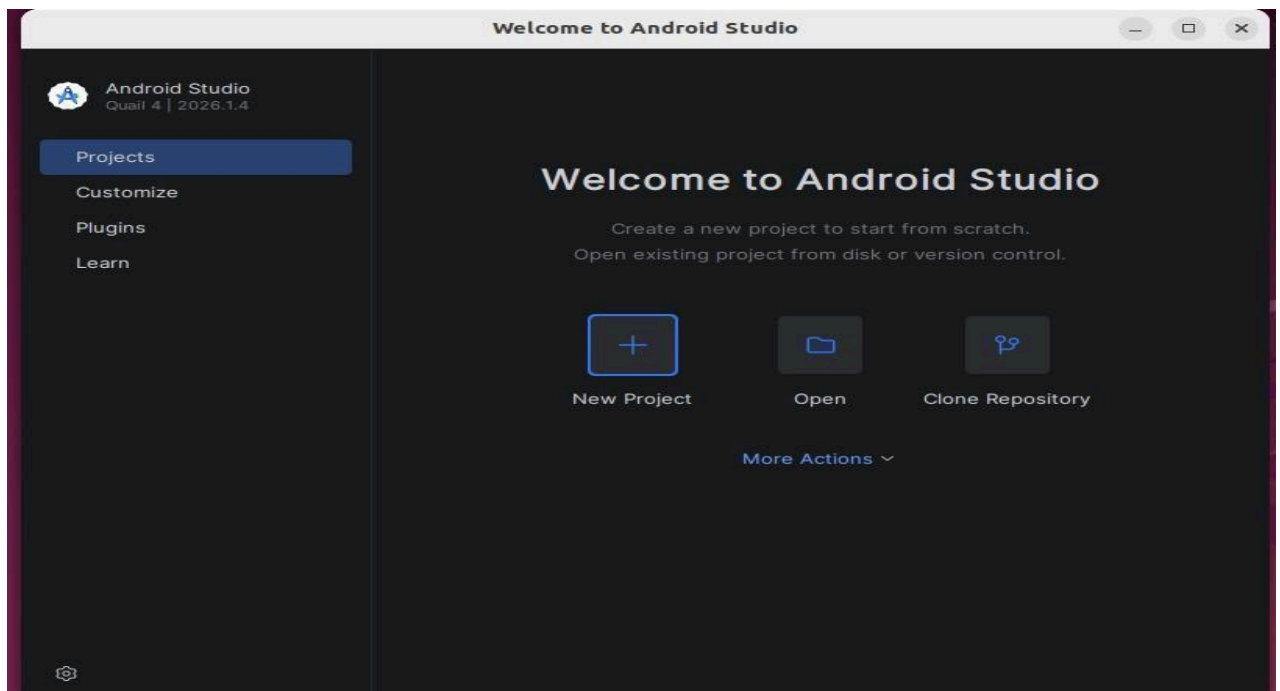
4.8 After this, the downloading and installation of the components get started.



4.9 Once all of the components have been downloaded, click on the **Finish** button.

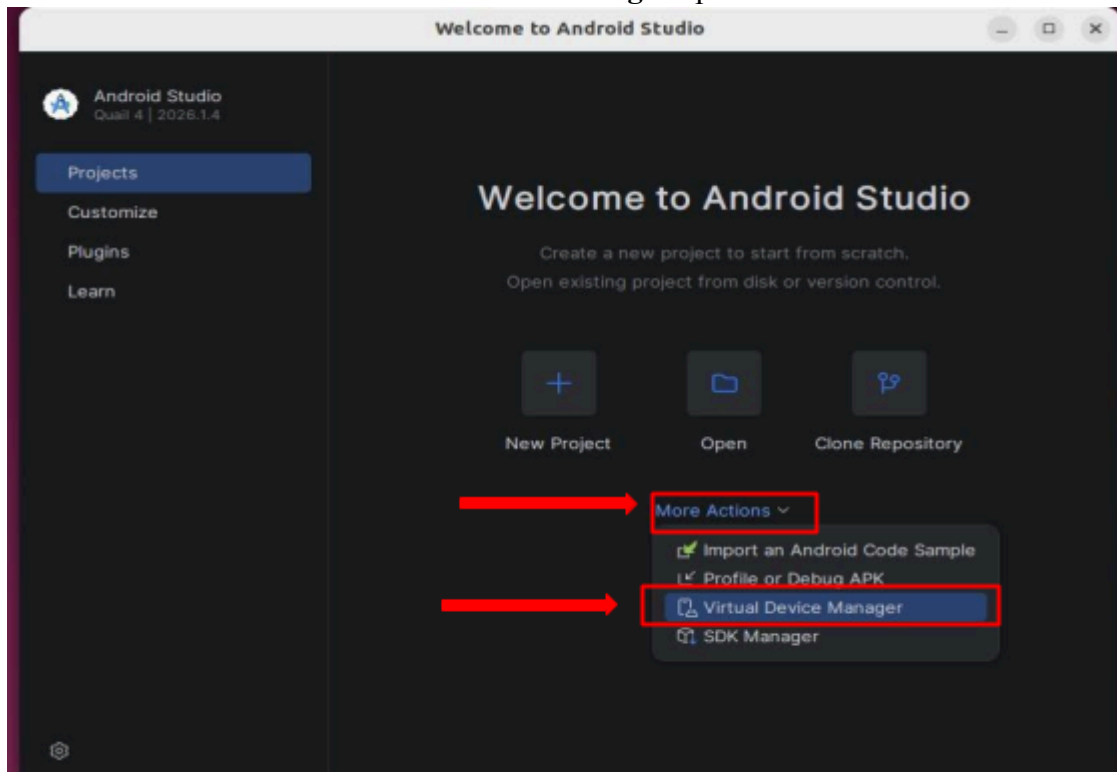


4.10 The setup of Android Studio IDE has been completed and is now ready to use.

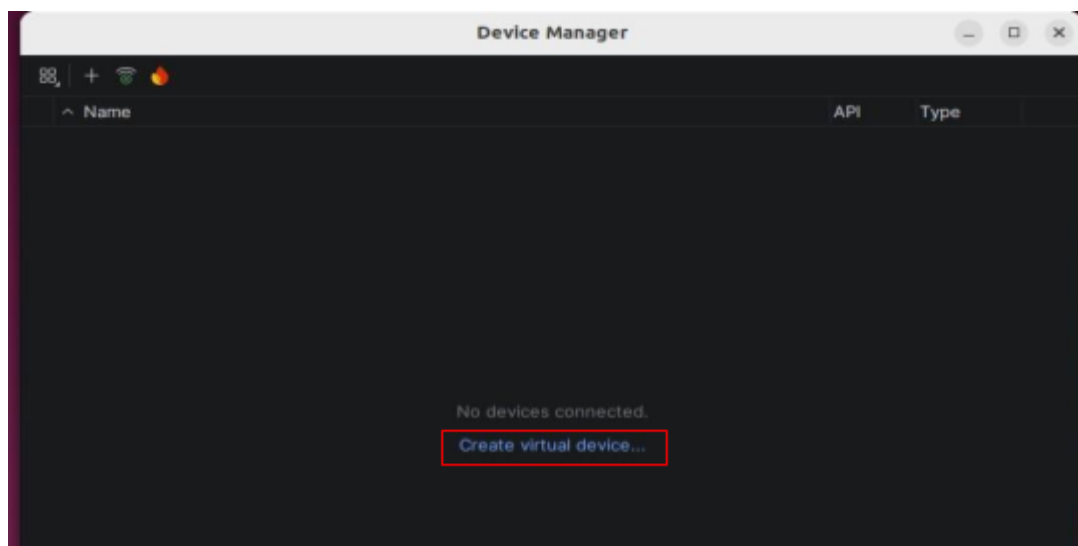


5. Configure Android Virtual Device (AVD)

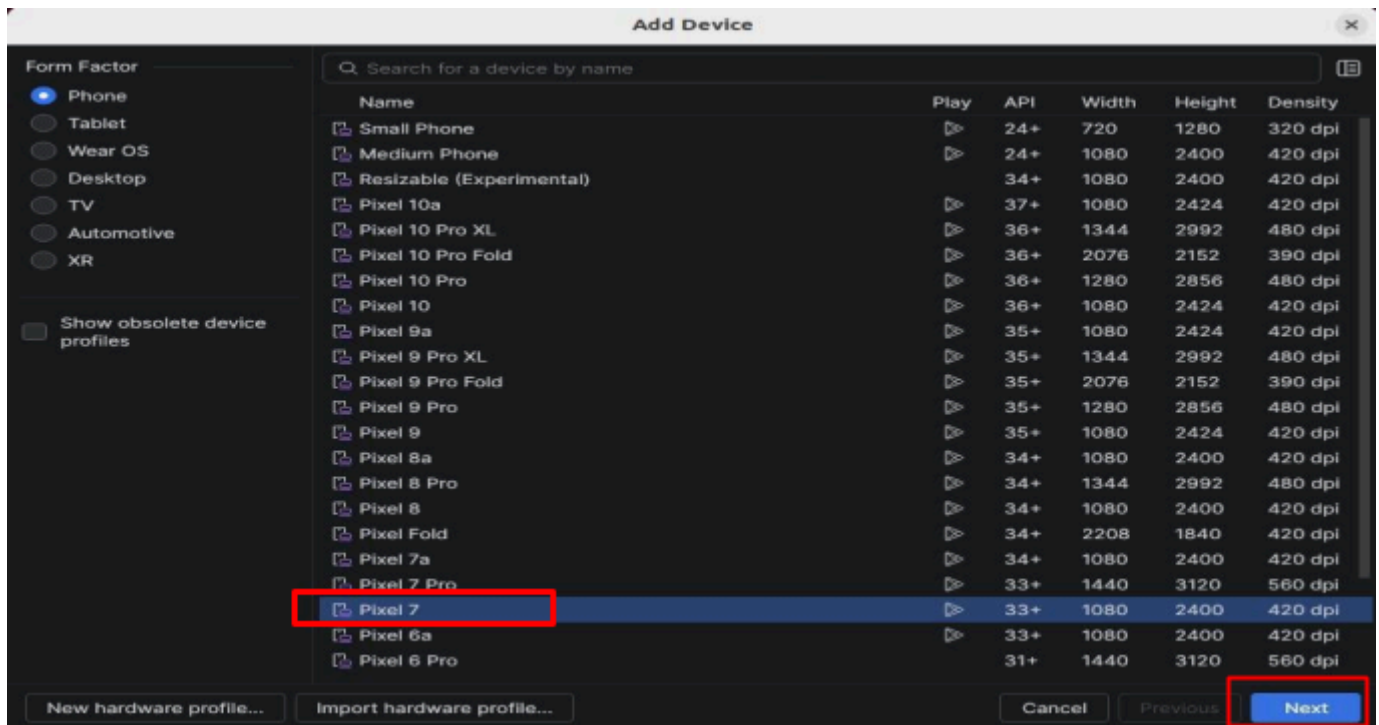
5.1 To set up the Android Virtual Device (AVD), open the Android Studio and then click on **More Actions** and select the **Virtual Device Manager** option.



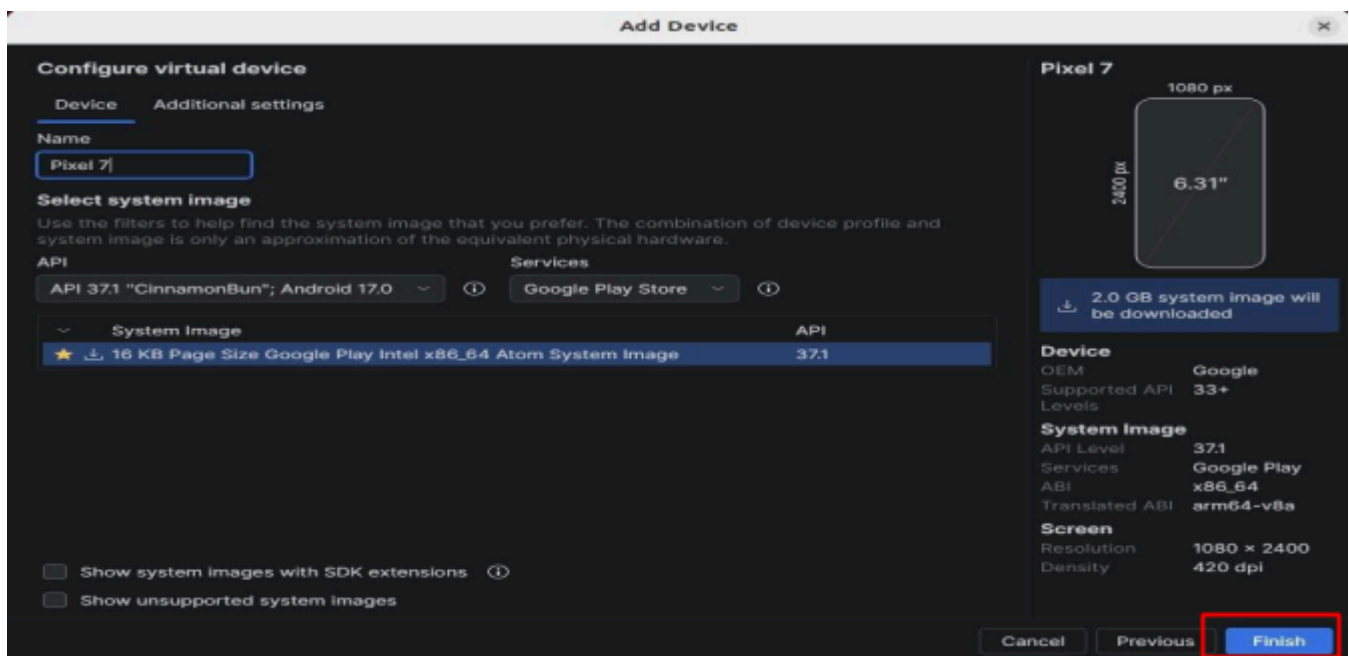
5.2 Click on the **Create Virtual Device** button and proceed.



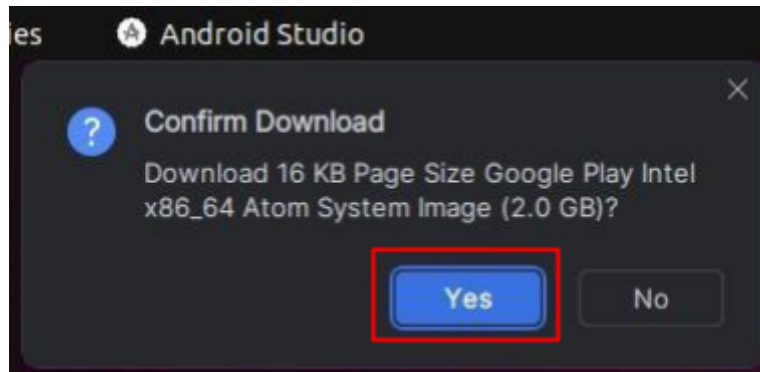
5.3 Select the device named **Pixel 7** from the available list of devices and then click on **Next** button.



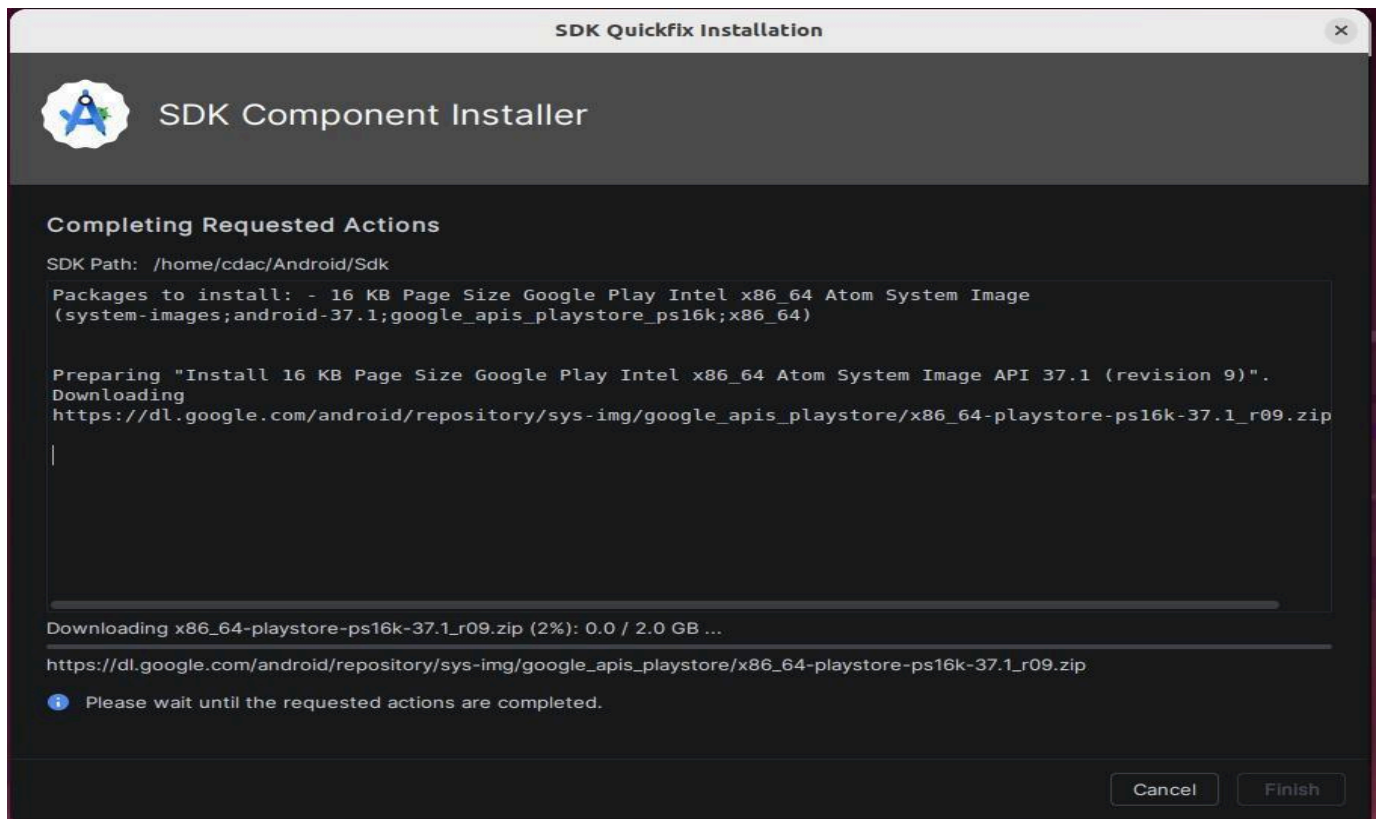
5.4 Now click on **Finish**



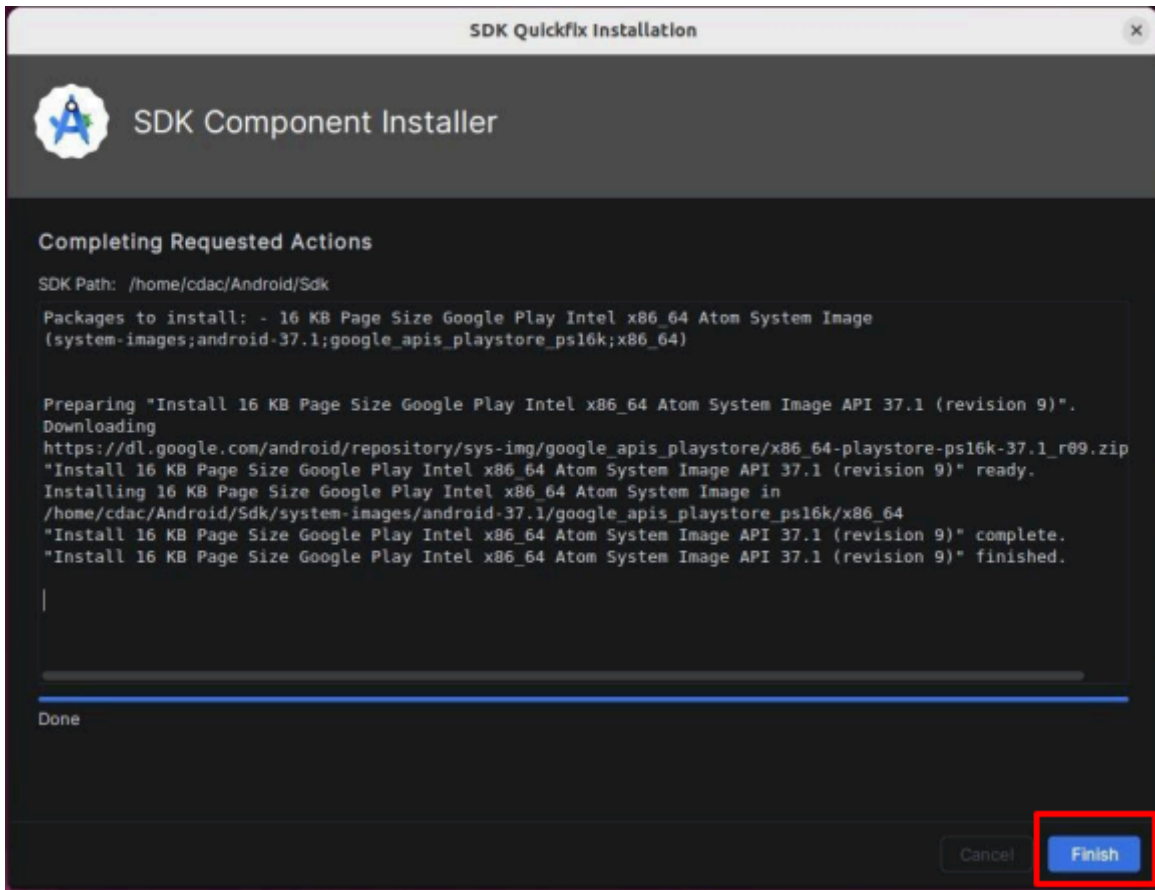
5.5 A Dialog Box Will Appear Click on Yes



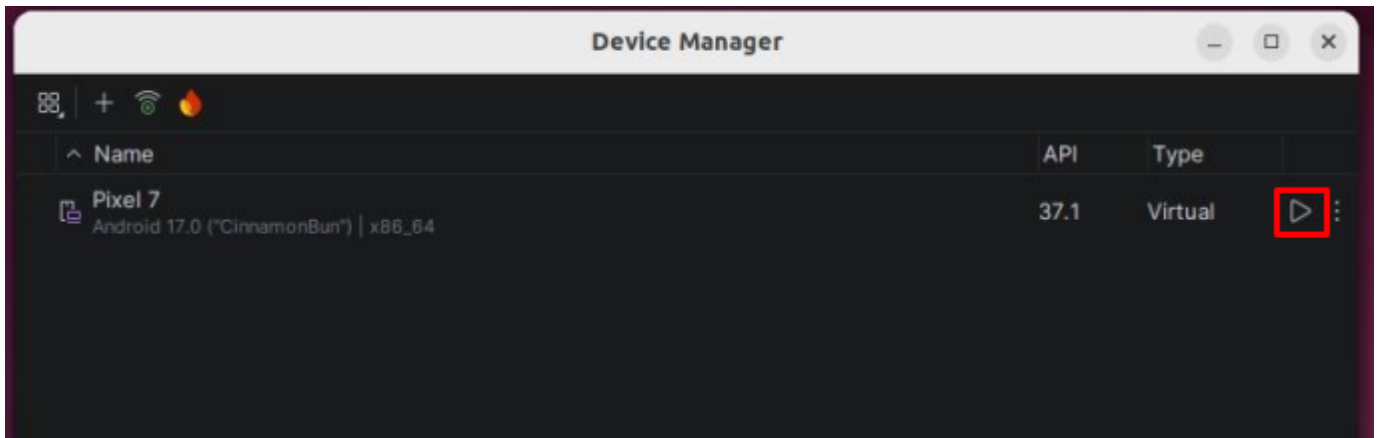
5.6 The downloading will get started.



5.7 Click on the **Finish** button once the downloading has been completed.



5.8 Finally select the Virtual Device created in above steps and then click on the ► (play) button and the emulator will start.



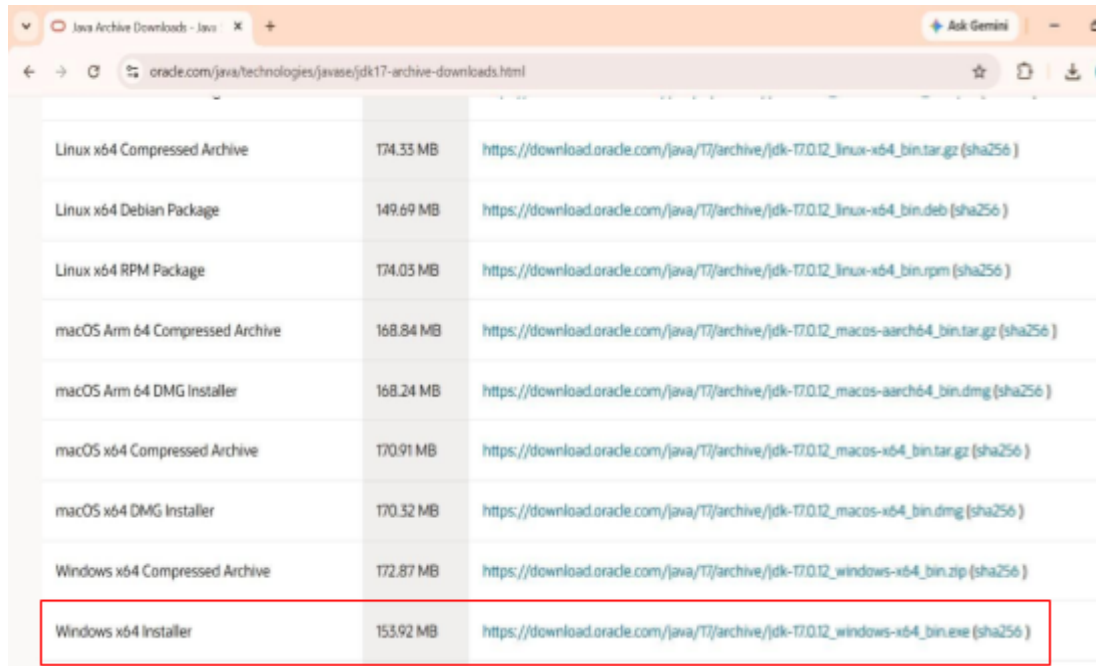
5.9 The emulator should look like this when started:



Step 2: Installing and Configuring on Windows System

1. Download Java Development Kit

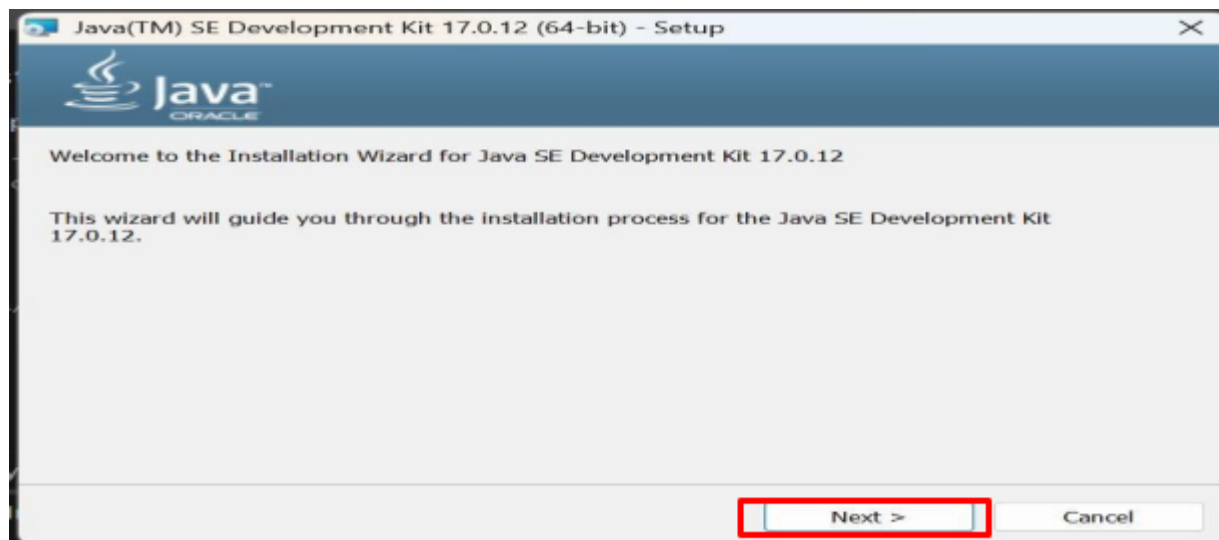
- 1.1. Go to <https://www.oracle.com/java/technologies/javase/jdk17-archive-downloads.html> website, scroll down and choose the Windows x64 Installer.



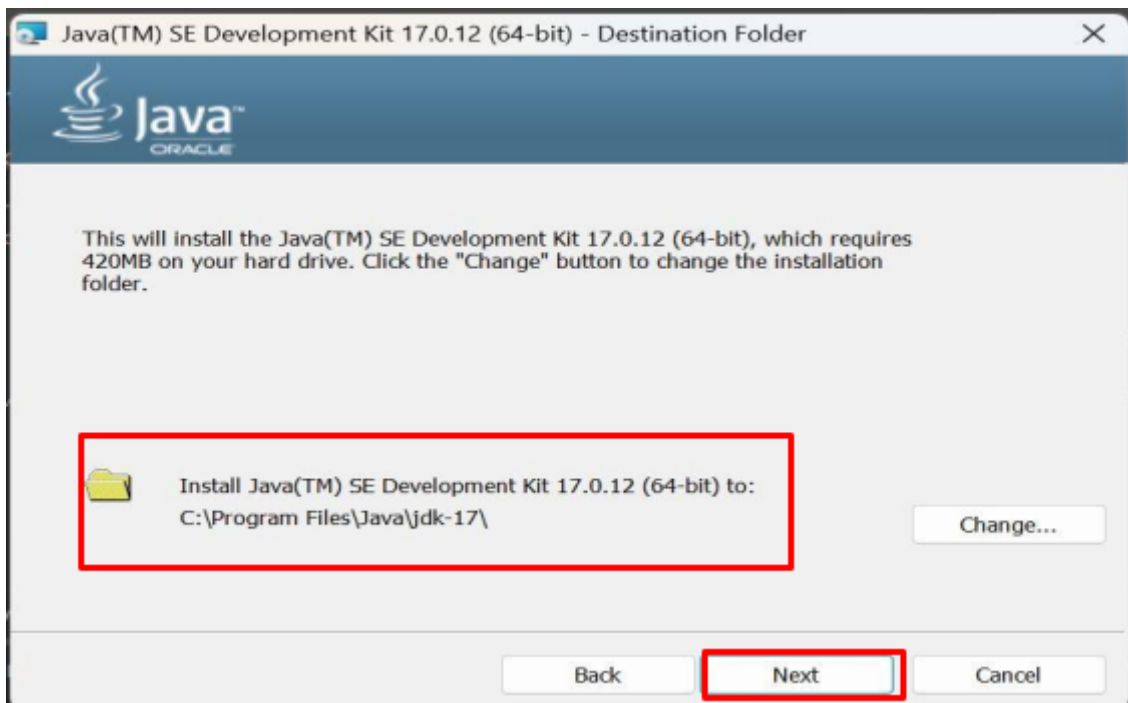
Linux x64 Compressed Archive	174.33 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_linux-x64_bin.tar.gz (sha256)
Linux x64 Debian Package	149.69 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_linux-x64_bin.deb (sha256)
Linux x64 RPM Package	174.05 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_linux-x64_bin.rpm (sha256)
macOS Arm 64 Compressed Archive	168.84 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-aarch64_bin.tar.gz (sha256)
macOS Arm 64 DMG Installer	168.24 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-aarch64_bin.dmg (sha256)
macOS x64 Compressed Archive	170.91 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-x64_bin.tar.gz (sha256)
macOS x64 DMG Installer	170.32 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-x64_bin.dmg (sha256)
Windows x64 Compressed Archive	172.87 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_windows-x64_bin.zip (sha256)
Windows x64 Installer	153.92 MB	https://download.oracle.com/java/17/archive/jdk-17.0.12_windows-x64_bin.exe (sha256)

2. Install the Java Development Kit

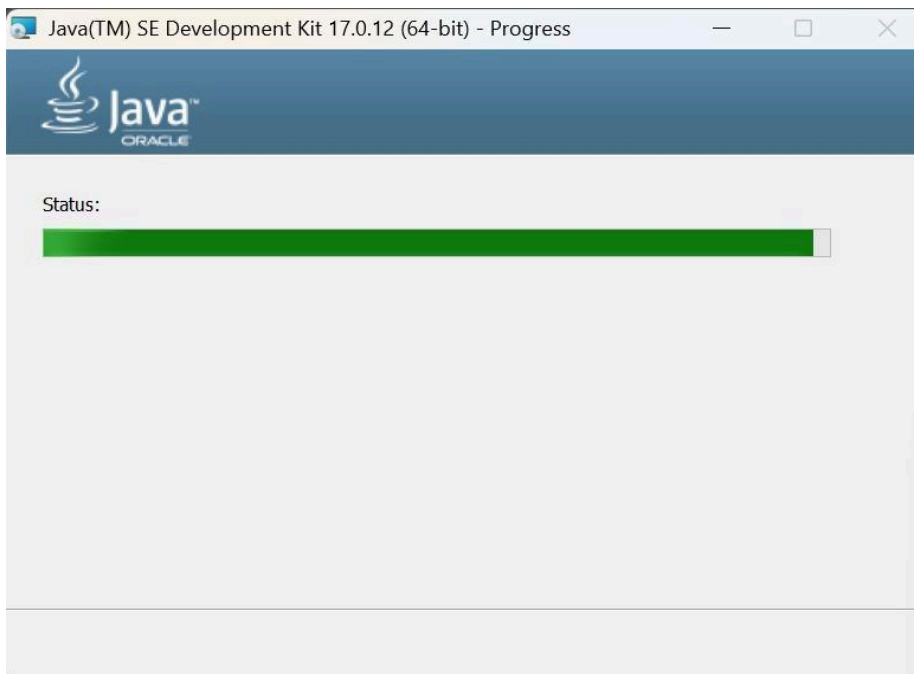
- 2.1. Once downloaded, open the .exe file and if prompted, then click on **Yes**.
- 2.2. Click on the **Next** button.



2.3. Leave the Installation Path to default and click on the **Next** button.



2.4. Installation will be in progress

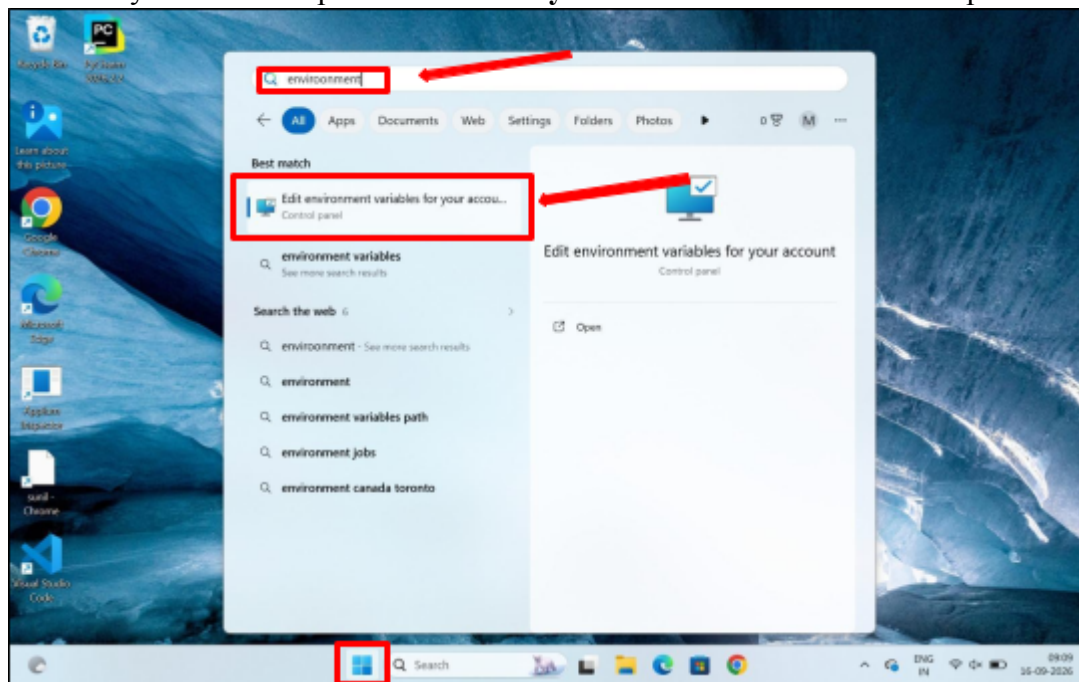


2.5. Once the installation has been completed, click on the **Close** button.

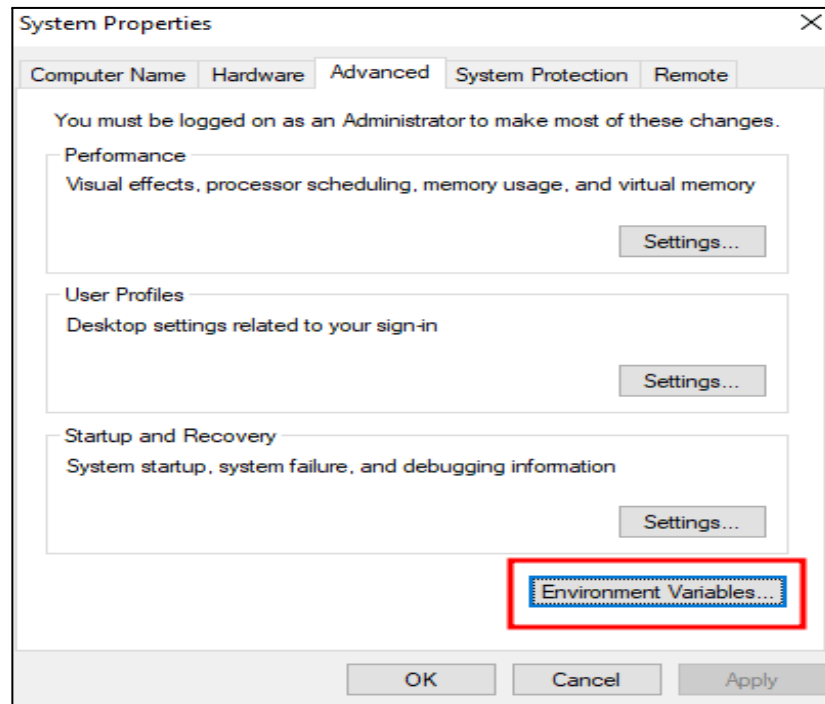


3. Configure the Java Environment

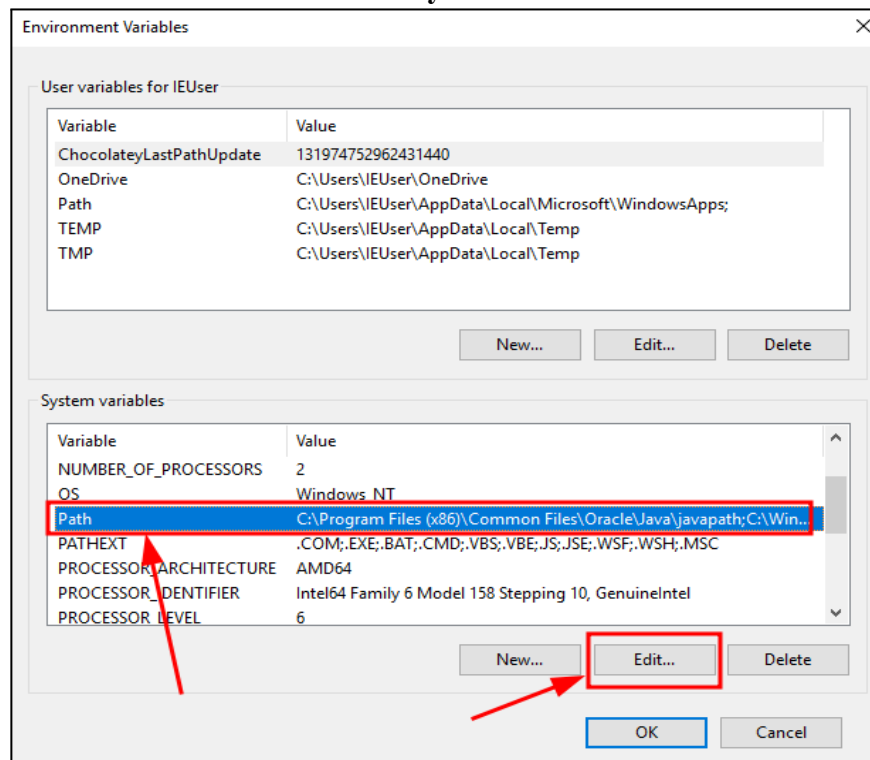
3.1. Click on the **Start** button located at the bottom left corner and search for the **environment** keyword. Then open the **Edit the system environment variables** option.



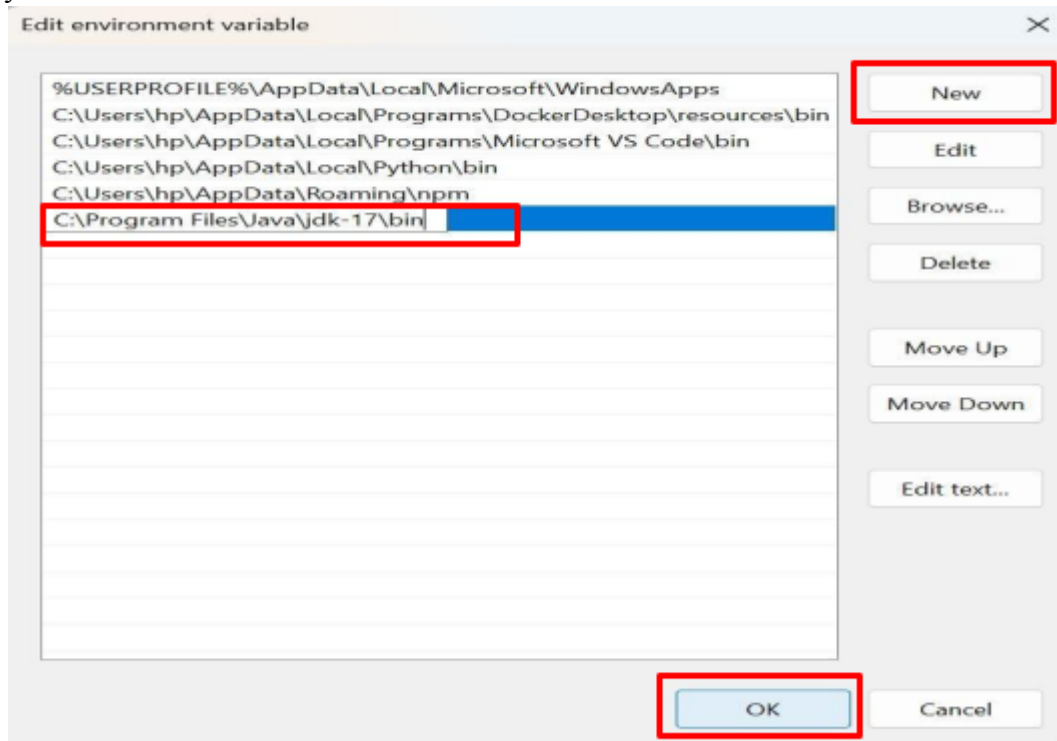
3.2. Click on **Environment Variables** to set the Java Runtime environment.



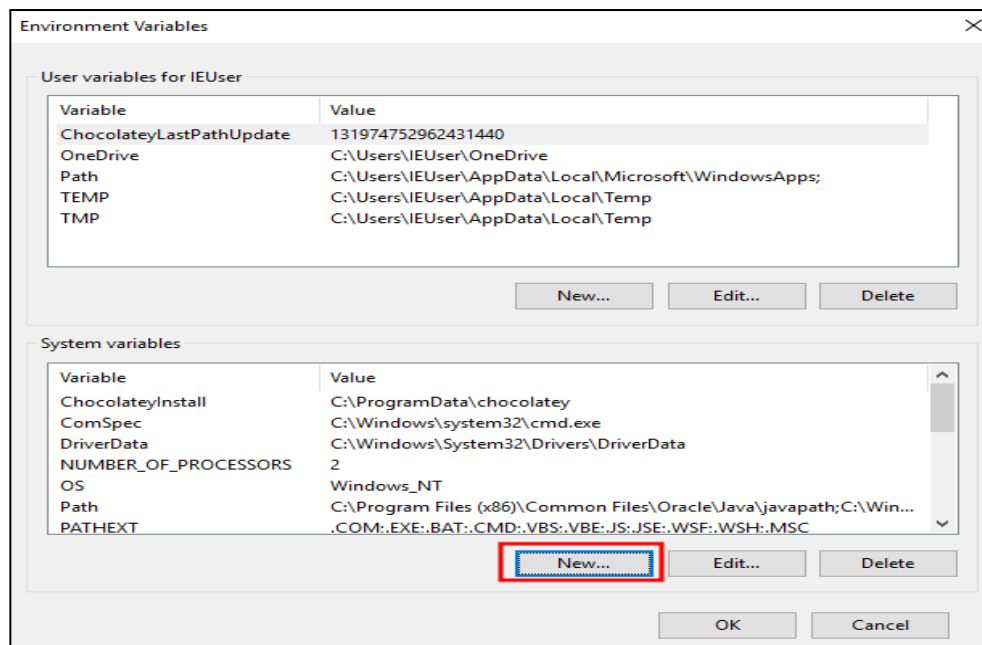
3.3. Select **Path** variable located under **System Variables** and then click on the **Edit** button.



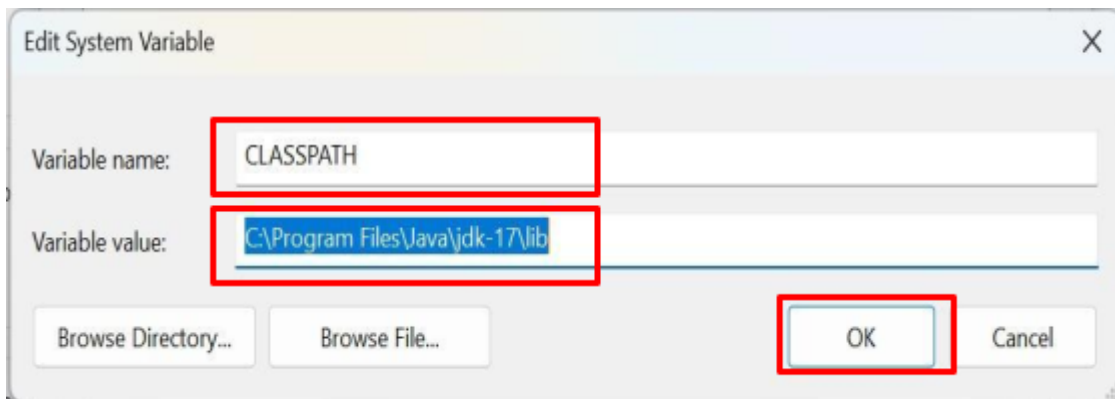
- 3.4. A new window will open. Click on the **New** button and write the path to JDK's bin folder where you installed the JDK and then click on the **OK** button.



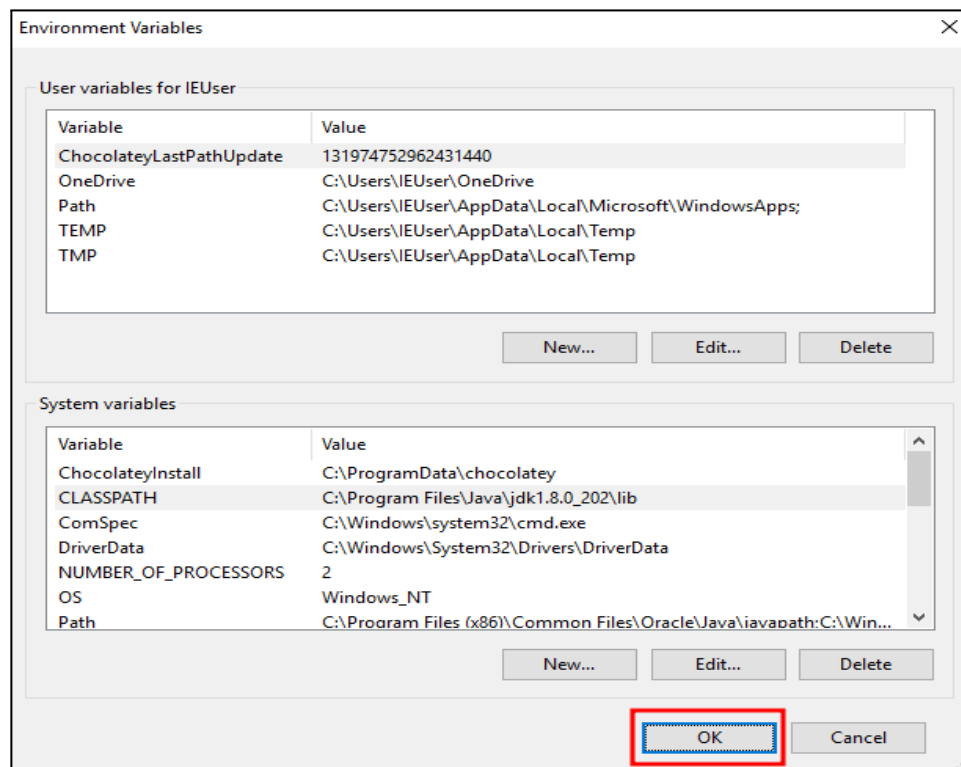
- 3.5. Now again click on the **New** button to add the **CLASSPATH** variable.



- 3.6. Enter **CLASSPATH** as Variable name and Path to JDK's **lib** directory as variable value and then click on **OK** button.

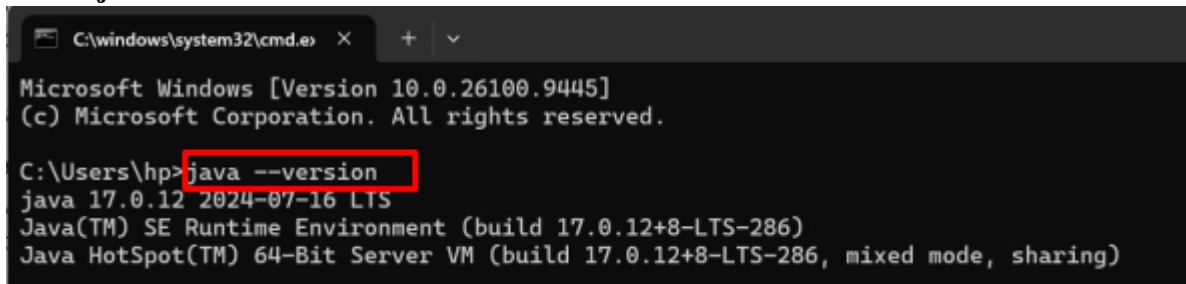


- 3.7. Finally click on the **OK** button.



- 3.8. Now open the **command prompt** and execute the following command. The command should output the current java version installed on the system confirming that java has been installed successfully.

java --version



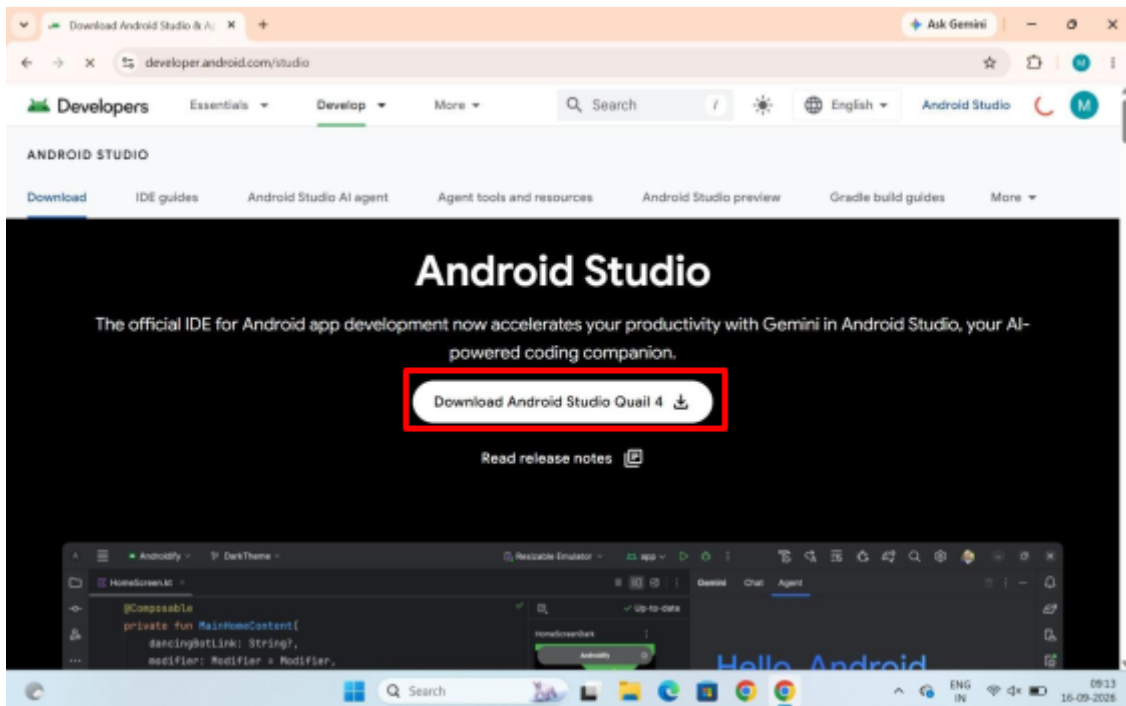
```
C:\windows\system32\cmd.exe X + v

Microsoft Windows [Version 10.0.26100.9445]
(c) Microsoft Corporation. All rights reserved.

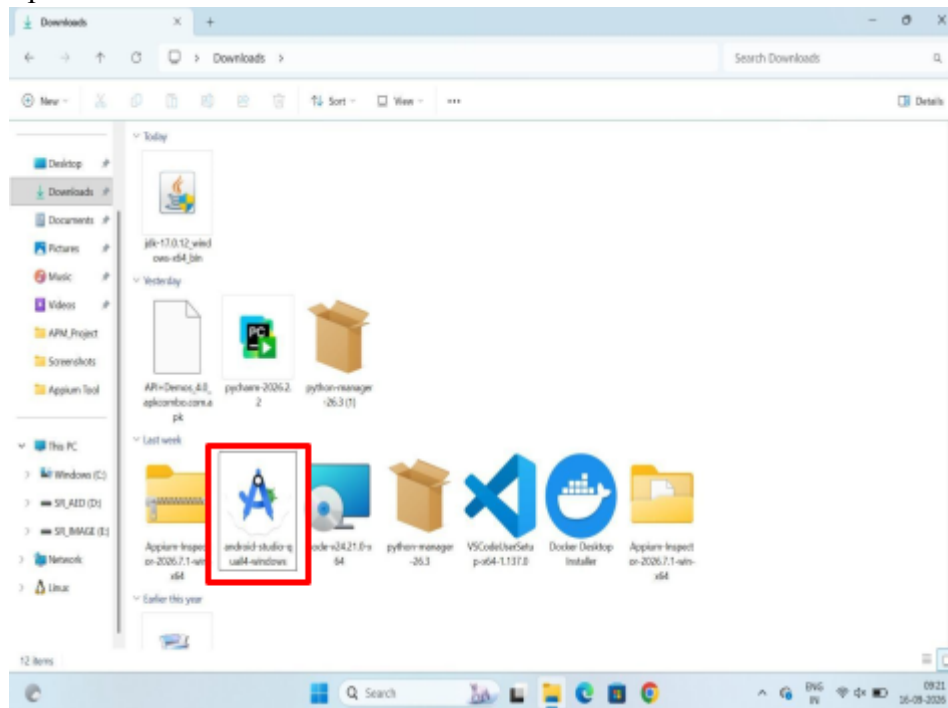
C:\Users\hp>java --version
java 17.0.12 2024-07-16 LTS
Java(TM) SE Runtime Environment (build 17.0.12+8-LTS-286)
Java HotSpot(TM) 64-Bit Server VM (build 17.0.12+8-LTS-286, mixed mode, sharing)
```

4. Download and Install Android Studio IDE

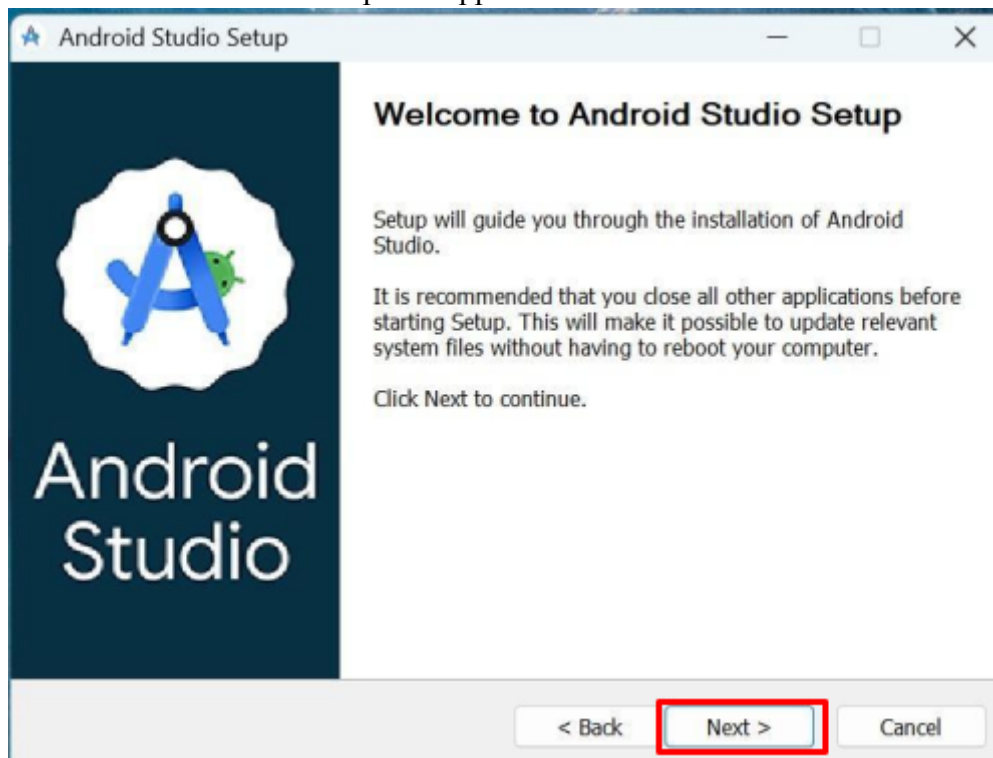
- 4.1. Go to <https://developer.android.com/studio> website and click on the **Download** Android Studio option.



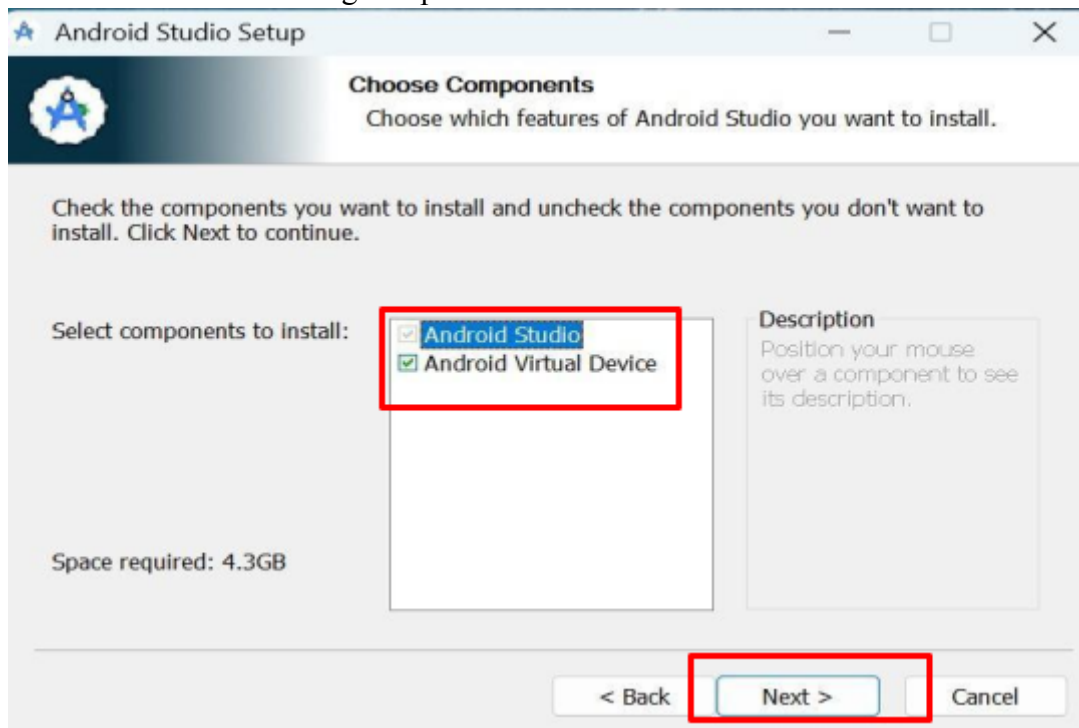
4.2. Open the downloaded **Android Studio.exe** file.



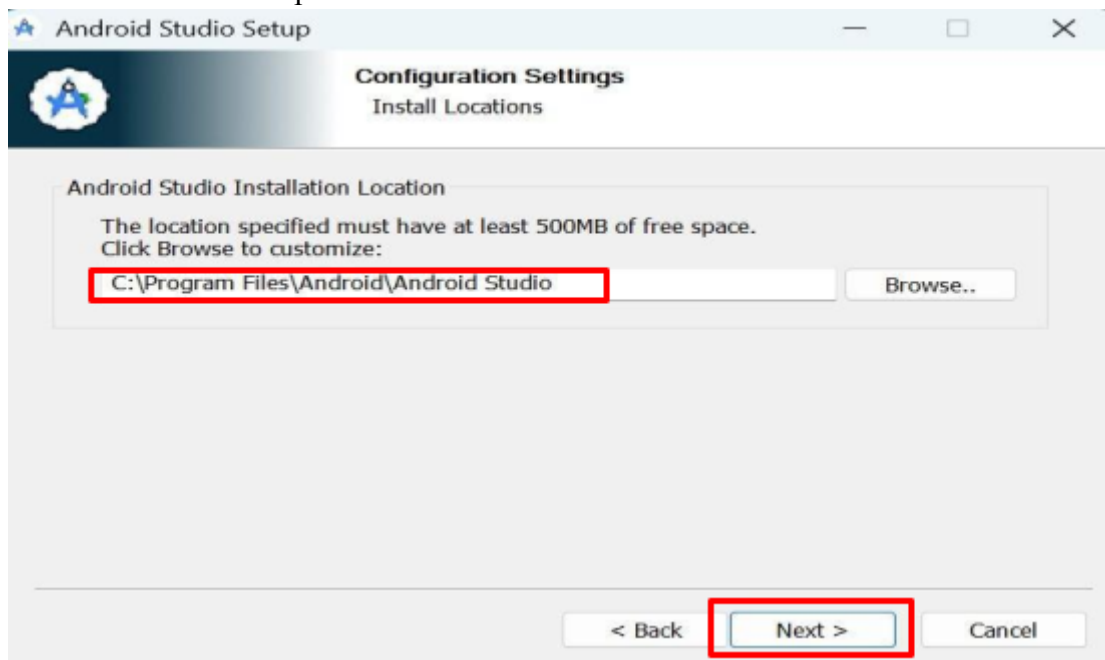
4.3. Android Studio Setup will appear on the screen and click the **Next** button to proceed.



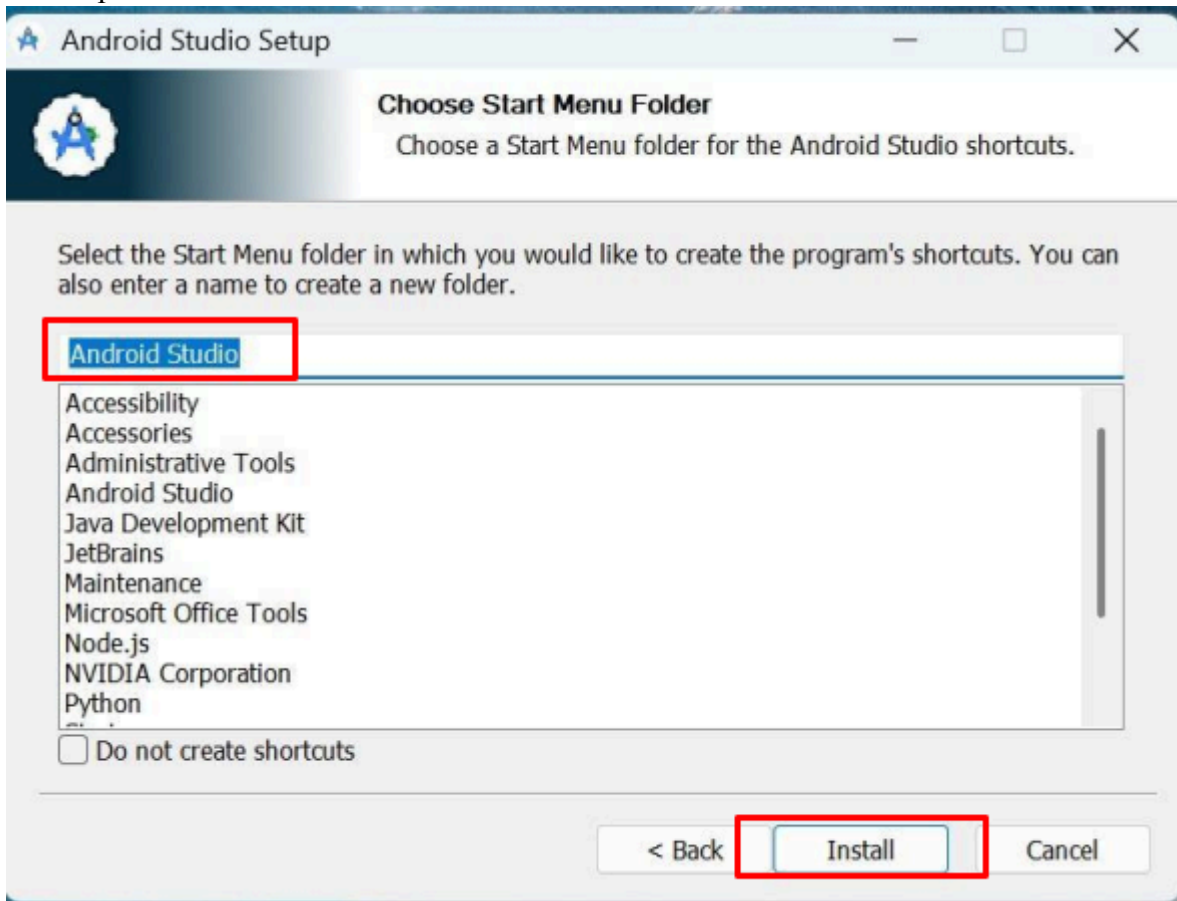
4.4. Select the following components and click on the **Next** button.



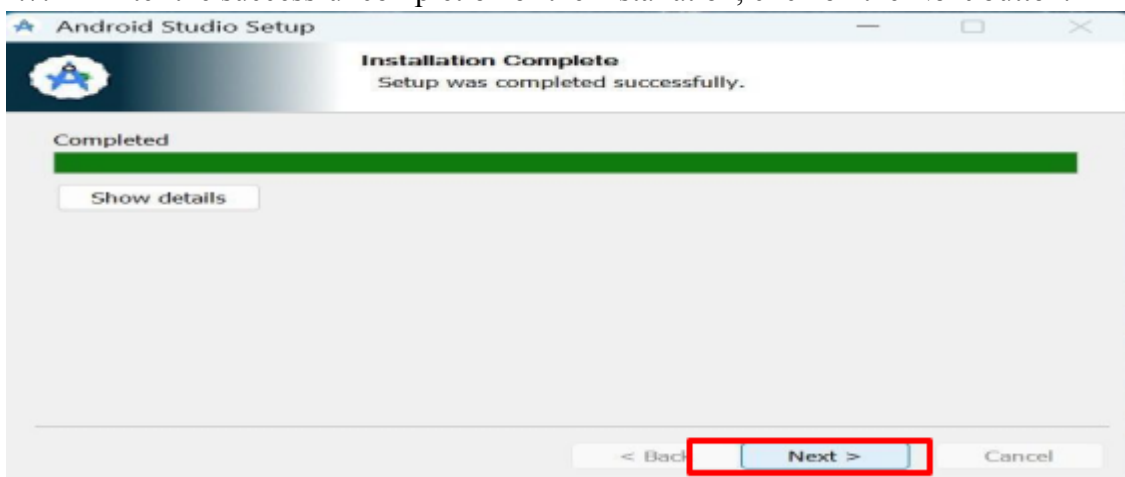
4.5. Now, browse the location where you want to install the Android Studio and click on the **Next** button to proceed.



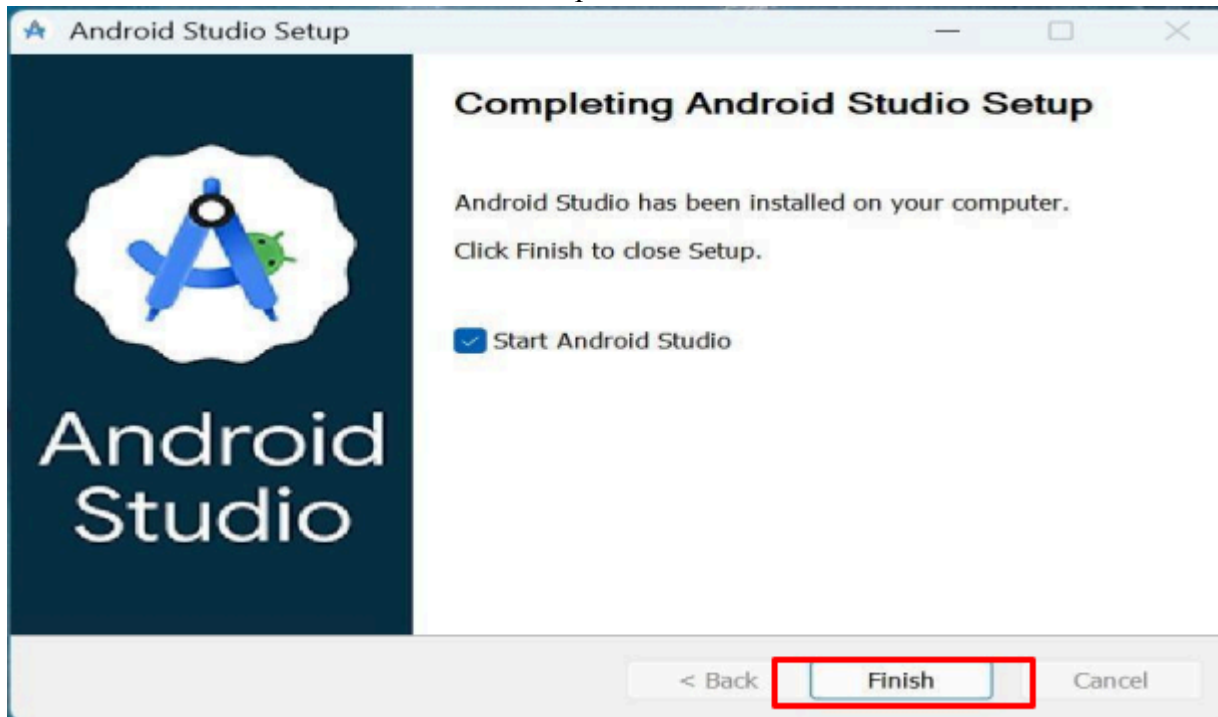
- 4.6. Choose a **start menu folder** for the Android Studio shortcut and click the **Install** button to proceed.



- 4.7. After the successful completion of the installation, click on the **Next** button.



4.8. Click on the **Finish** button to proceed.

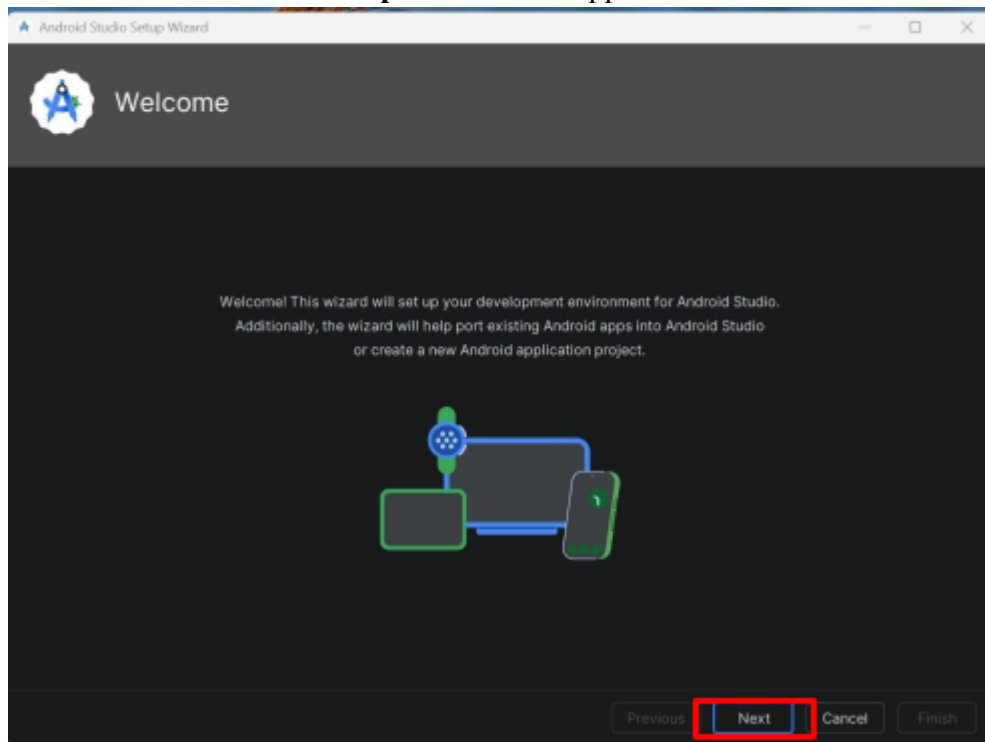


5. Configure the Android Studio IDE

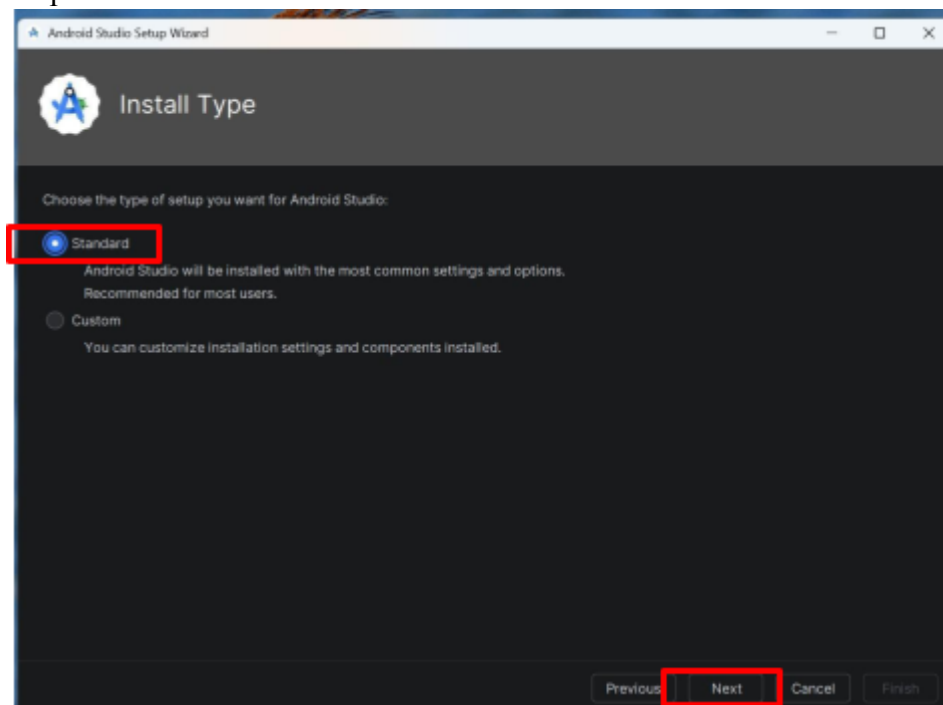
5.1. Once the Android Studio installation has been finished completely, the welcome screen will appear automatically like the following.



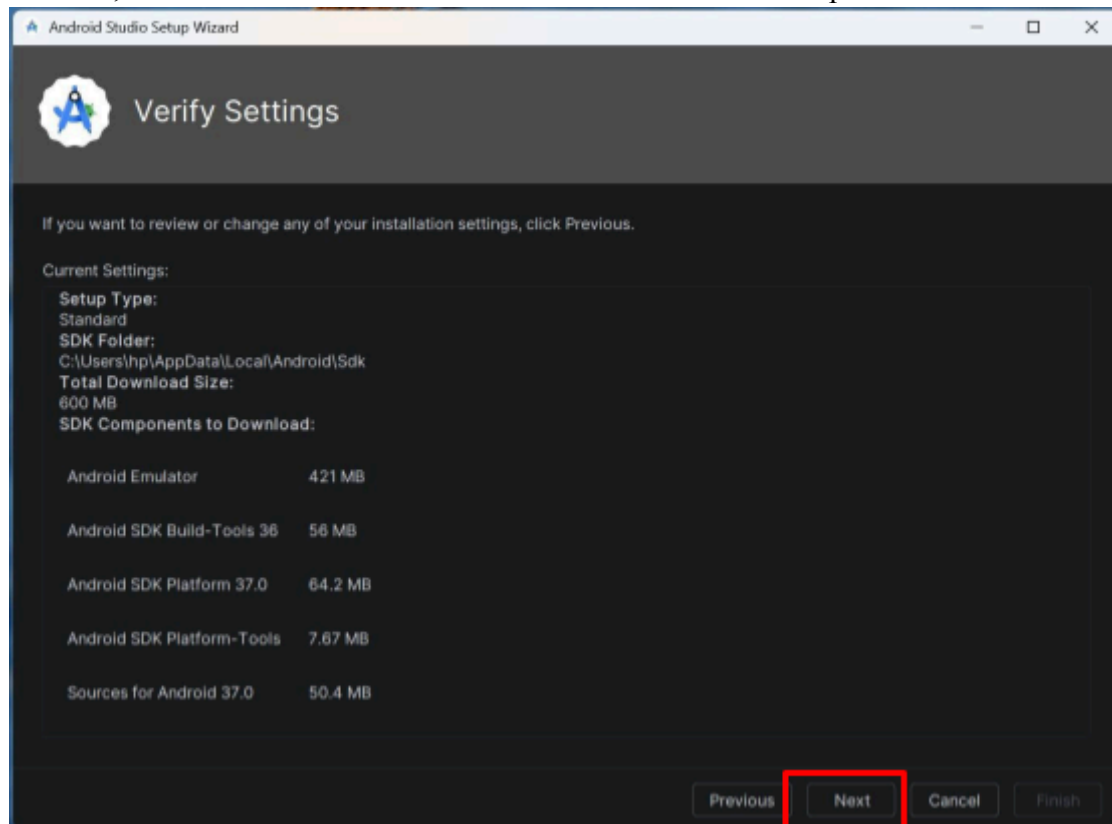
5.2. **Android Studio Setup Wizard** will appear on the screen. Click on the **Next** button



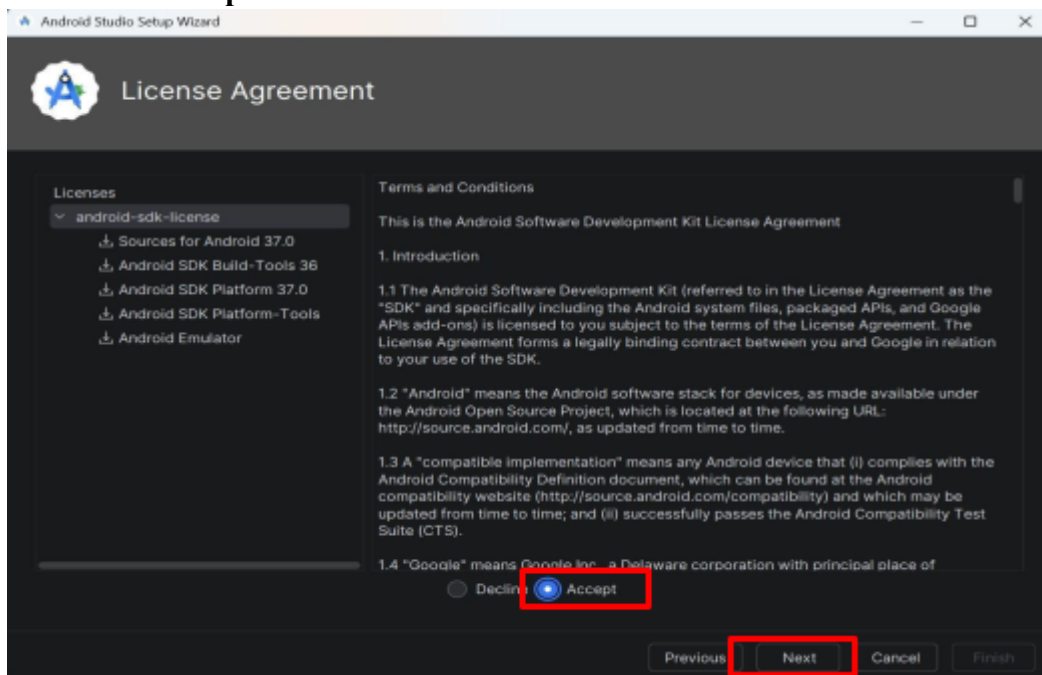
5.3. Now, select (check) the **Standard** option as installation type and click the **Next** button to proceed.



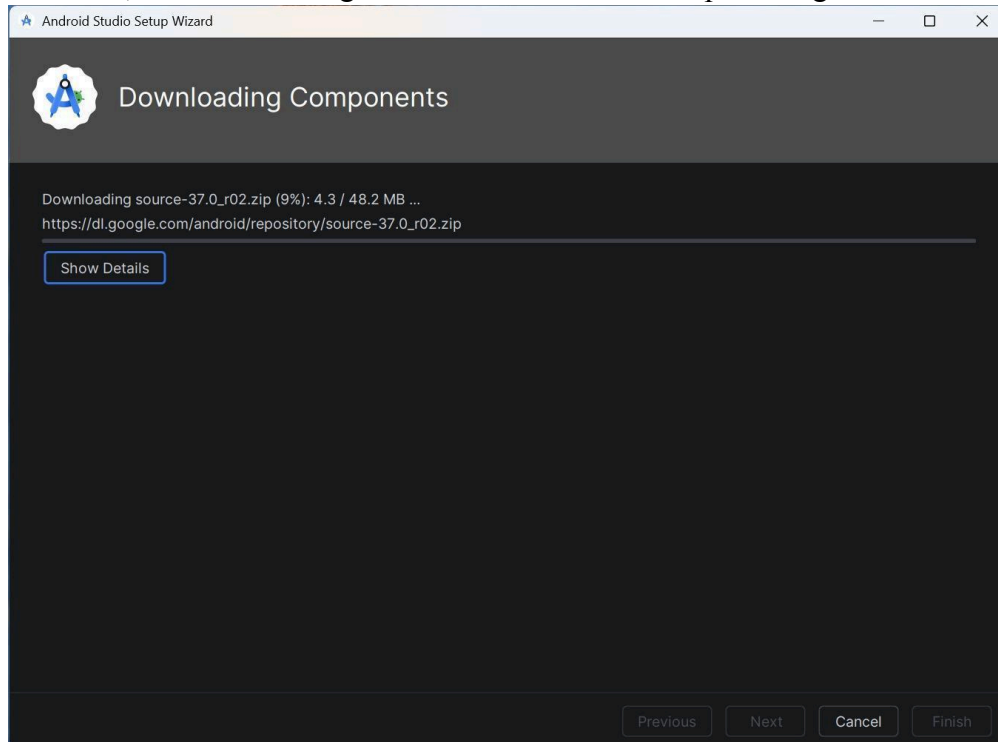
5.4. Now, click on the **Next** button to download all the SDK components.



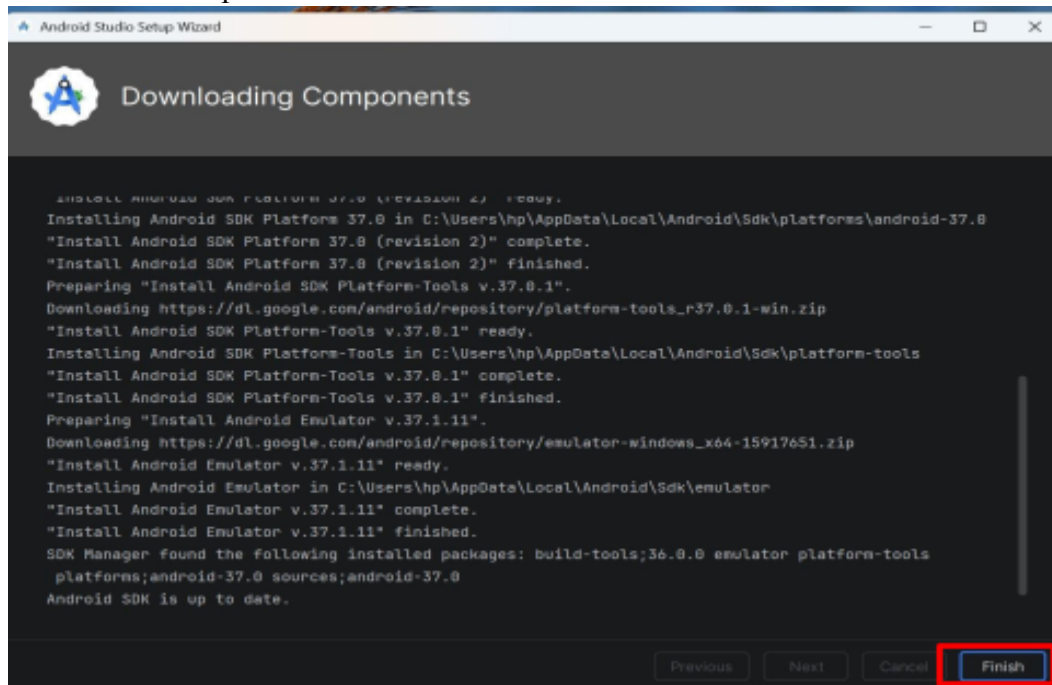
5.5. Now Select **Accept** and Click on **Next**



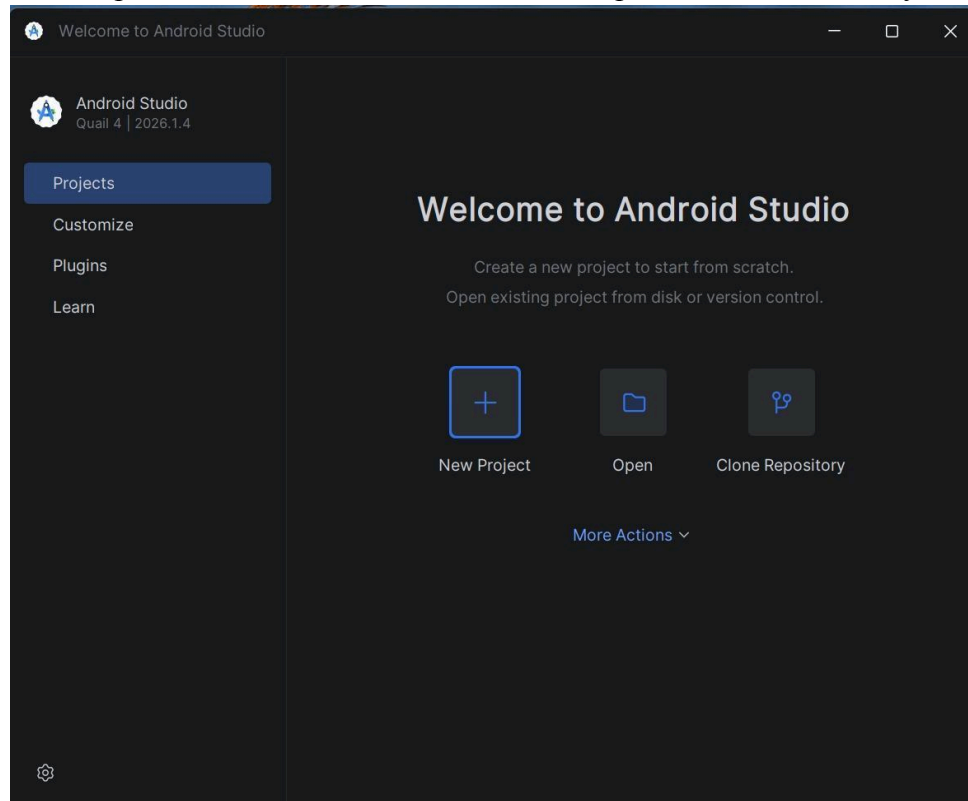
- 5.6. After this, the downloading and installation of the components get started.



- 5.7. Once all the components have been downloaded click on the **Finish** Button

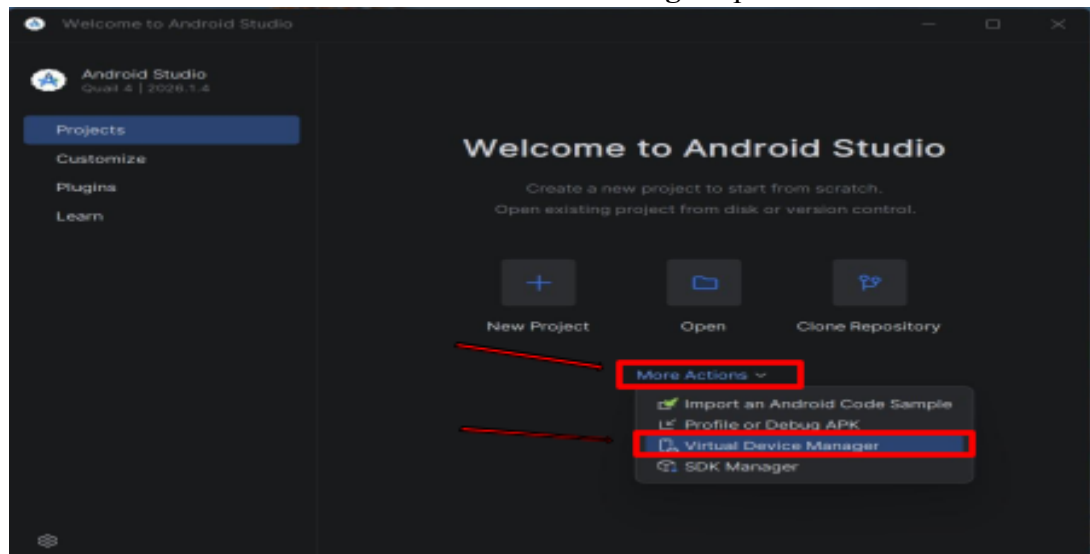


5.8. The setup of Android Studio IDE has been completed and is now ready to use.

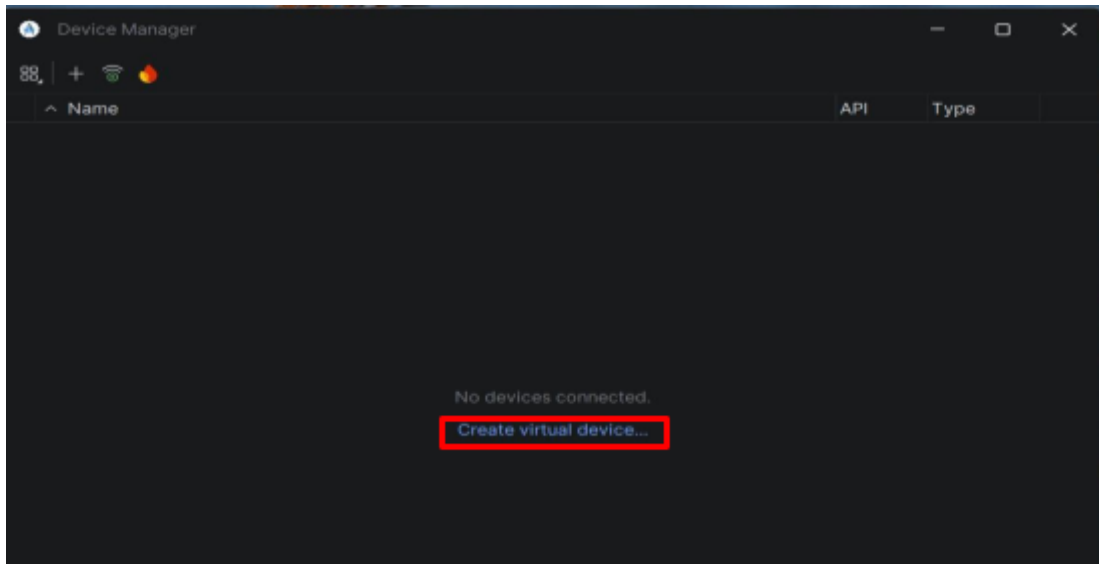


6. Configure Android Virtual Device (AVD)

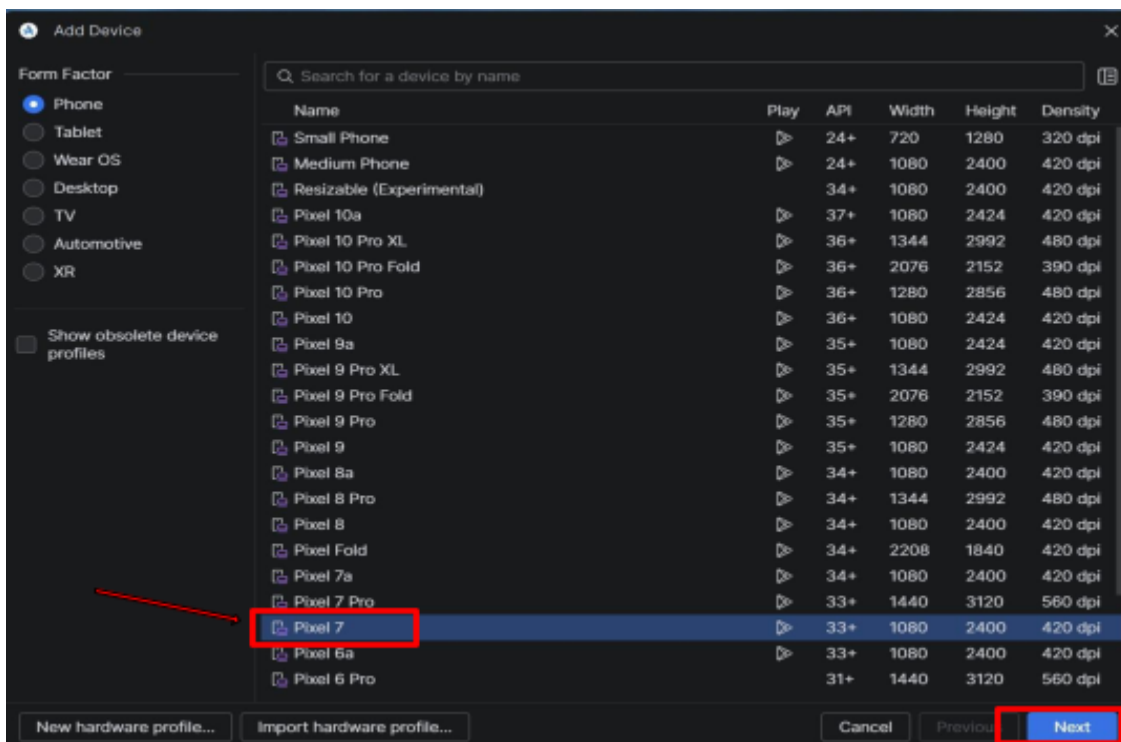
6.1. To set up the Android Virtual Device (AVD), open the Android Studio and then click on **More Actions** and select the **Virtual Device Manager** option.



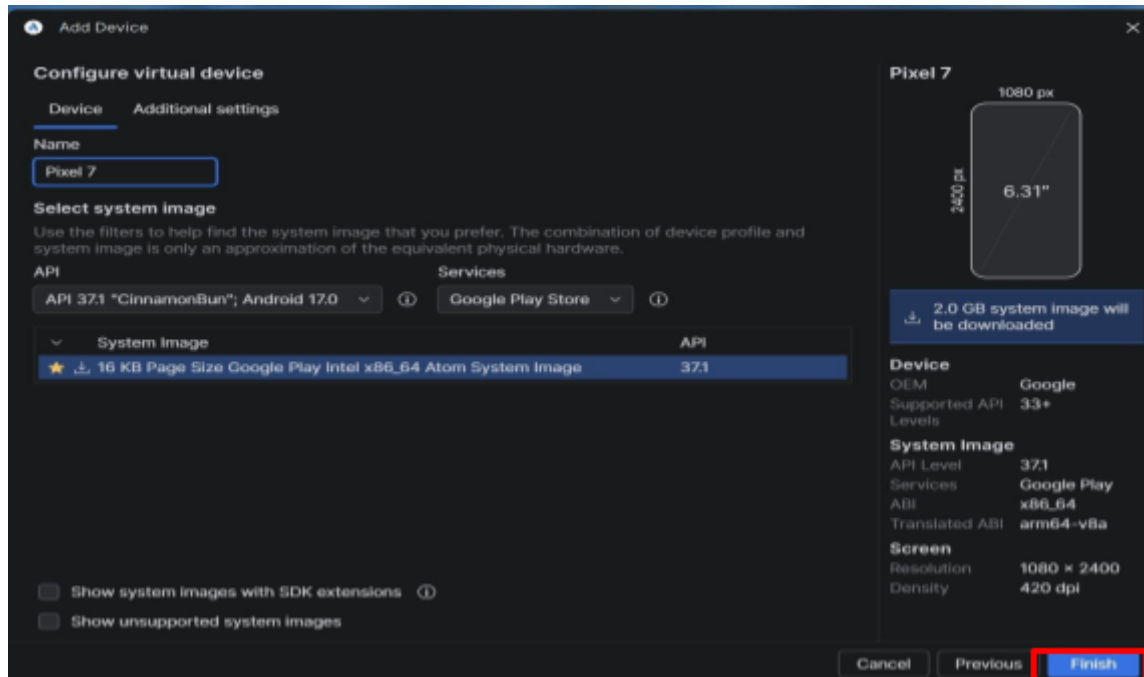
6.2. Click on the **Create Virtual Device** button and proceed.



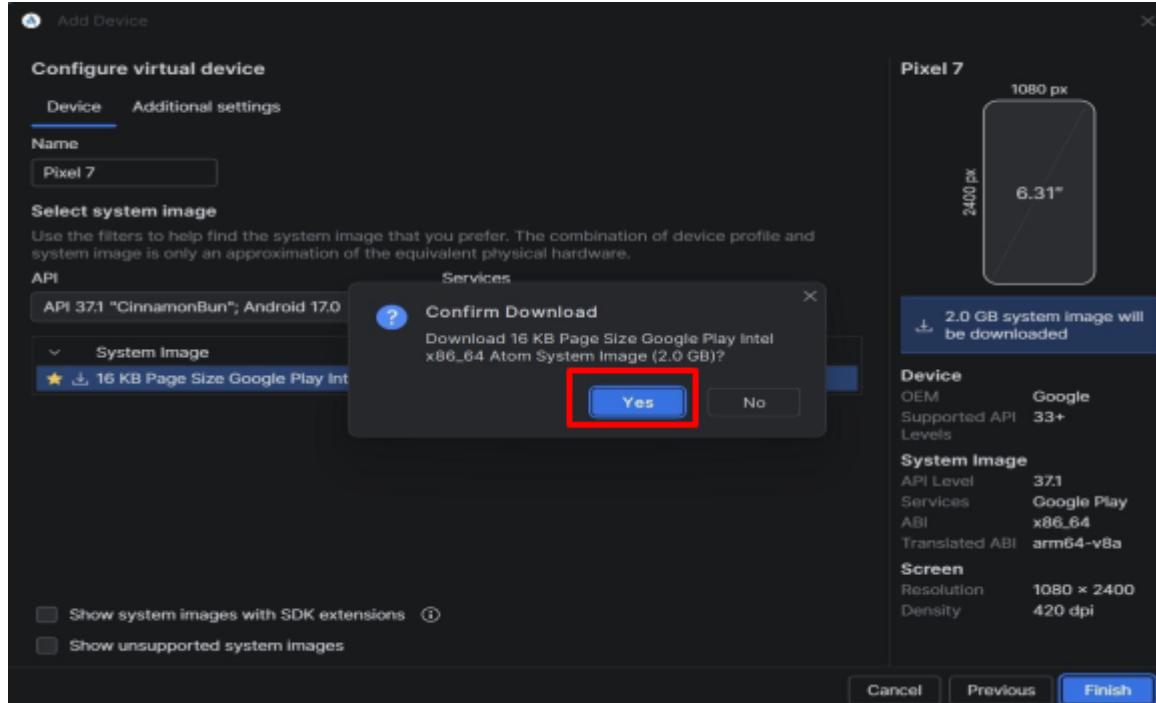
6.3. Select the device named **Pixel 7** from the available list of devices and then click on the **Next** button.



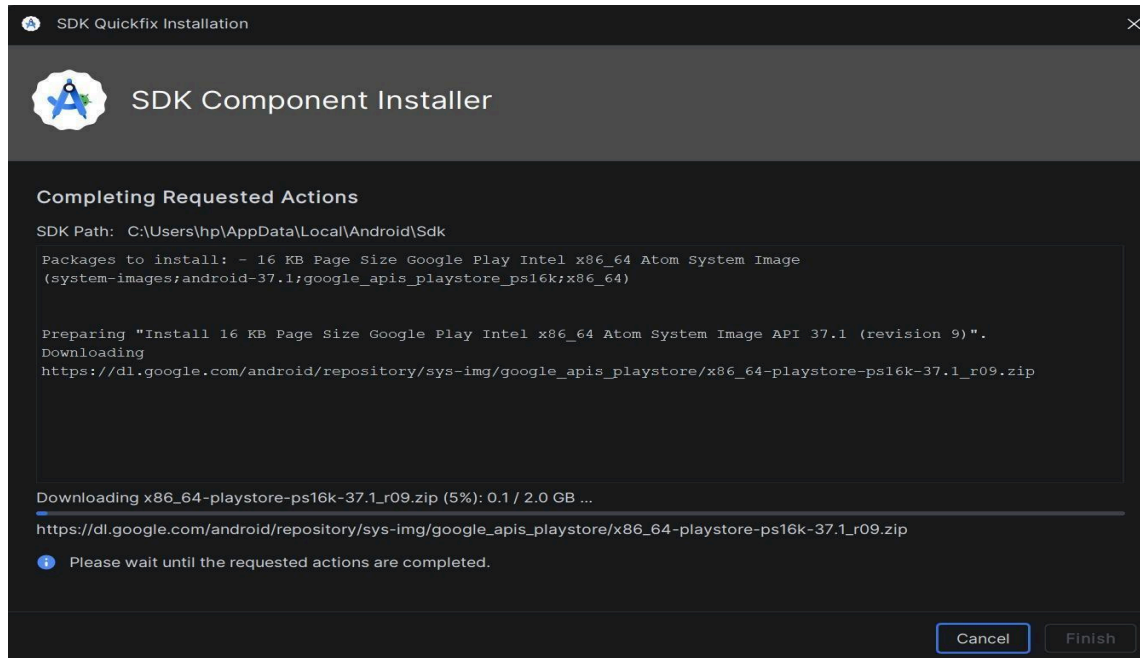
6.4. Now Click on **Finish**



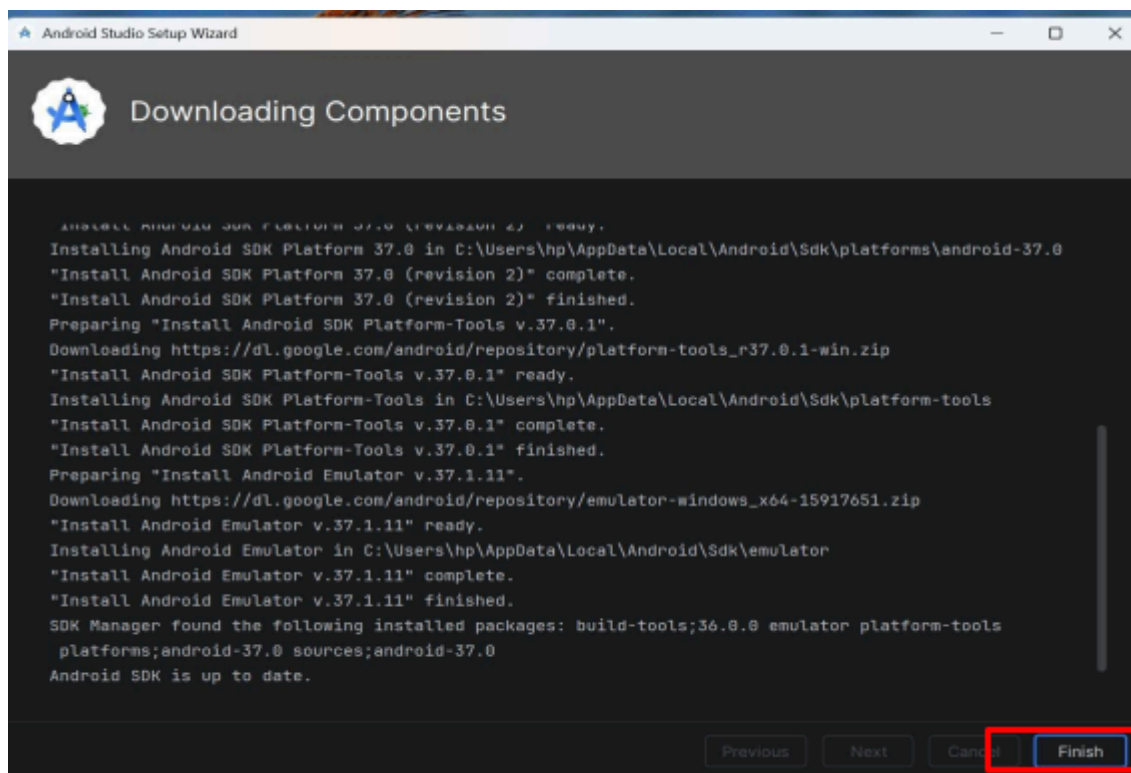
6.5. A Confirm Dialog Box will appear Click on **Yes**.



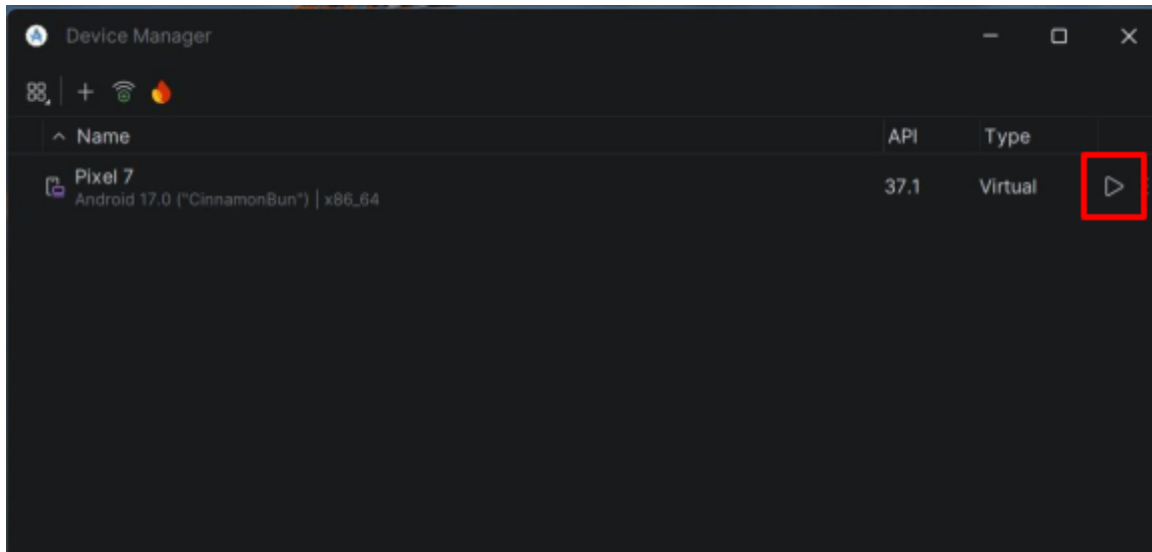
6.6. After Clicking on **Yes** The downloading will get started



6.7. After Successful download click on **Finish**



- 6.8. Finally select the Virtual Device created in above steps and then click on the ► (play) button and the emulator will start.

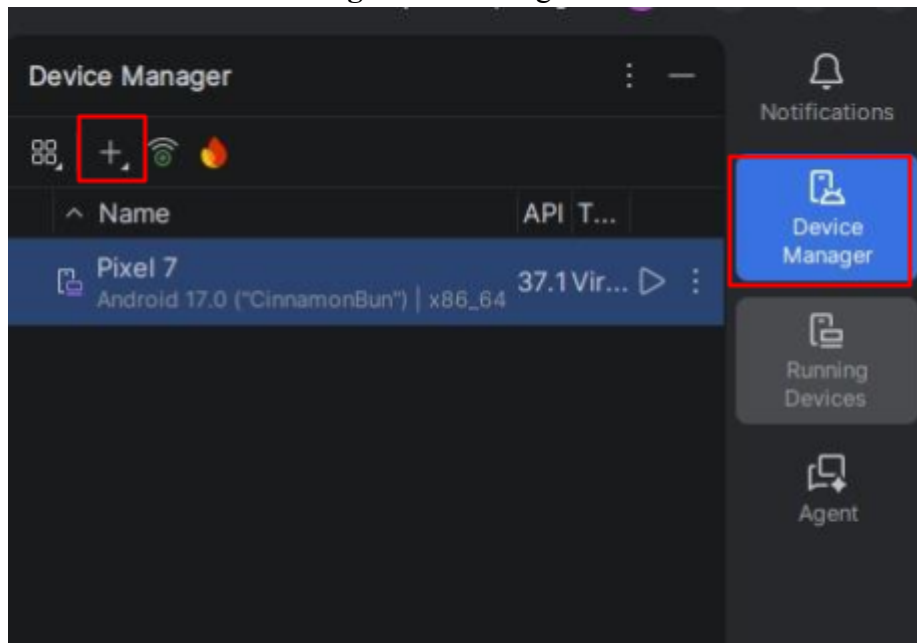


- 6.9. The emulator should look like this when started:

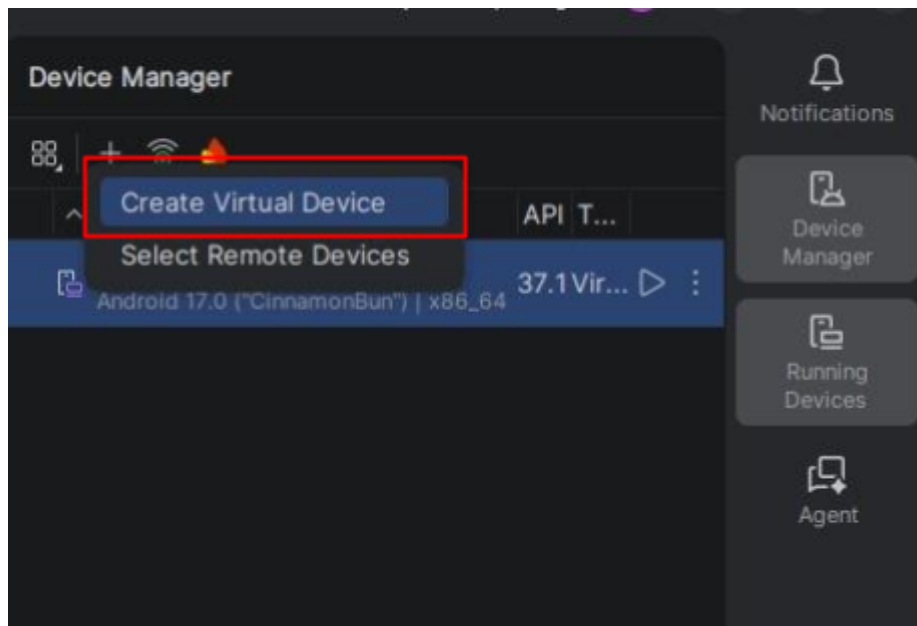


7. Installing a Non-Play Store Emulator

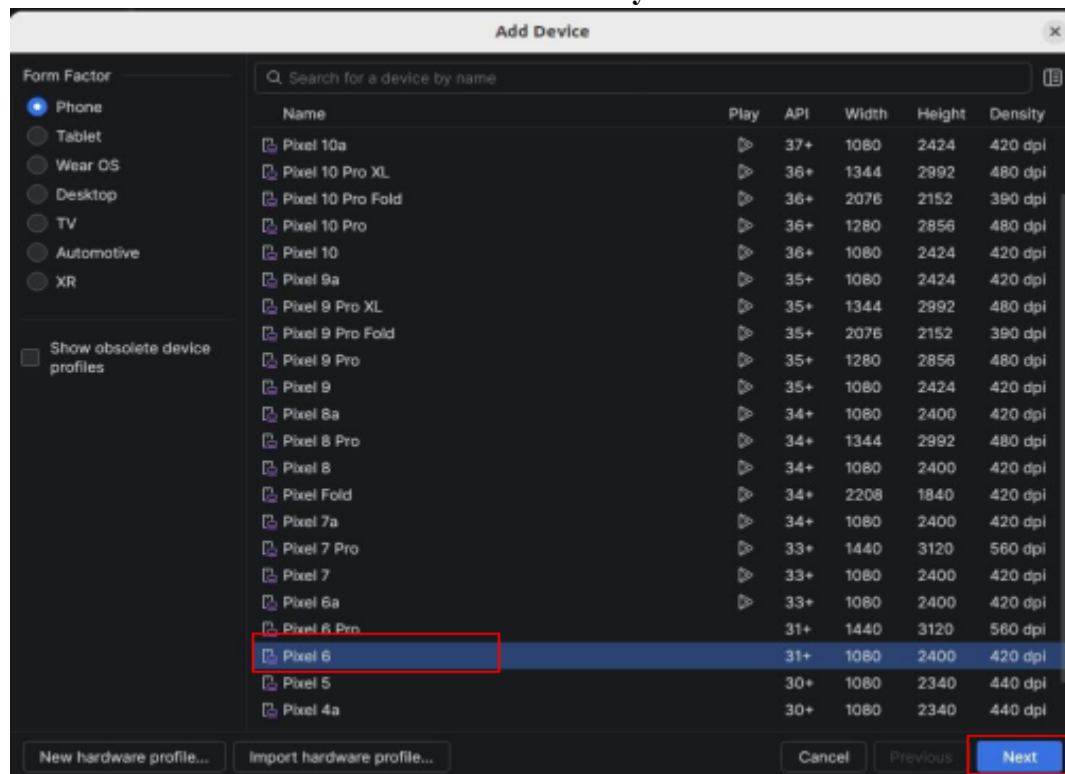
7.1. Click on the **Device Manager** from the right side of the screen and click on the “+” Button.



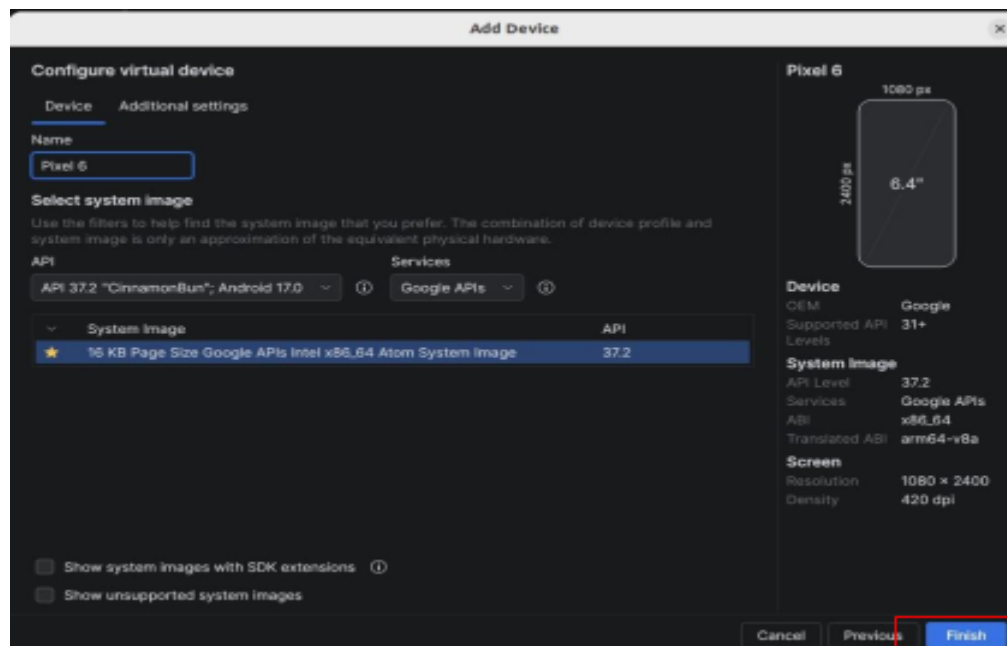
7.2. Now Click on **Create Virtual Device**.



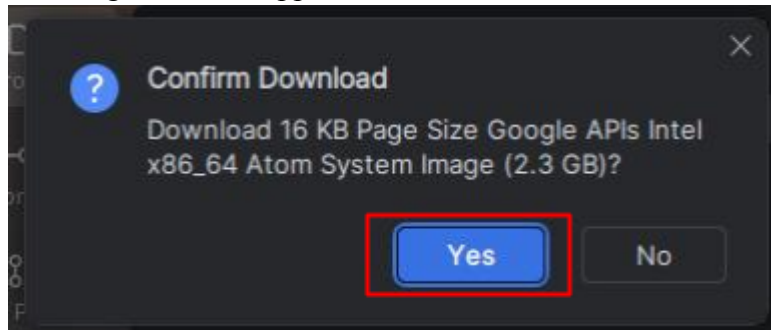
7.3. Now from the list select **Pixel 6** Without the **Play Store Icon** and Click on **Next**.



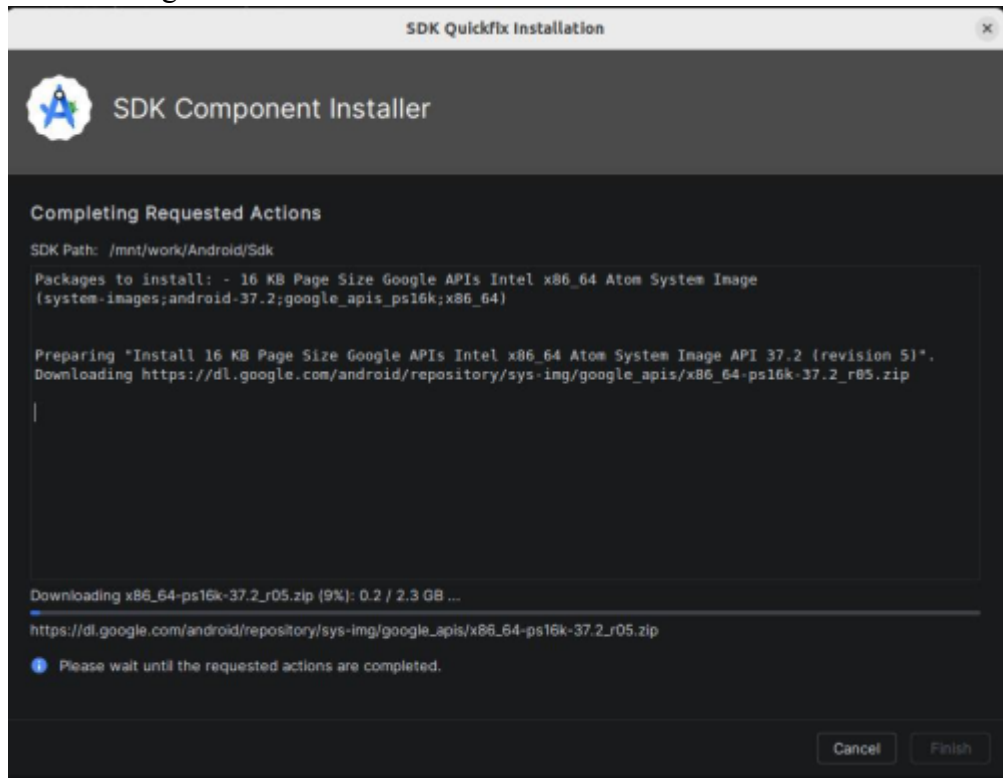
7.4. Now Click on **Finish**.



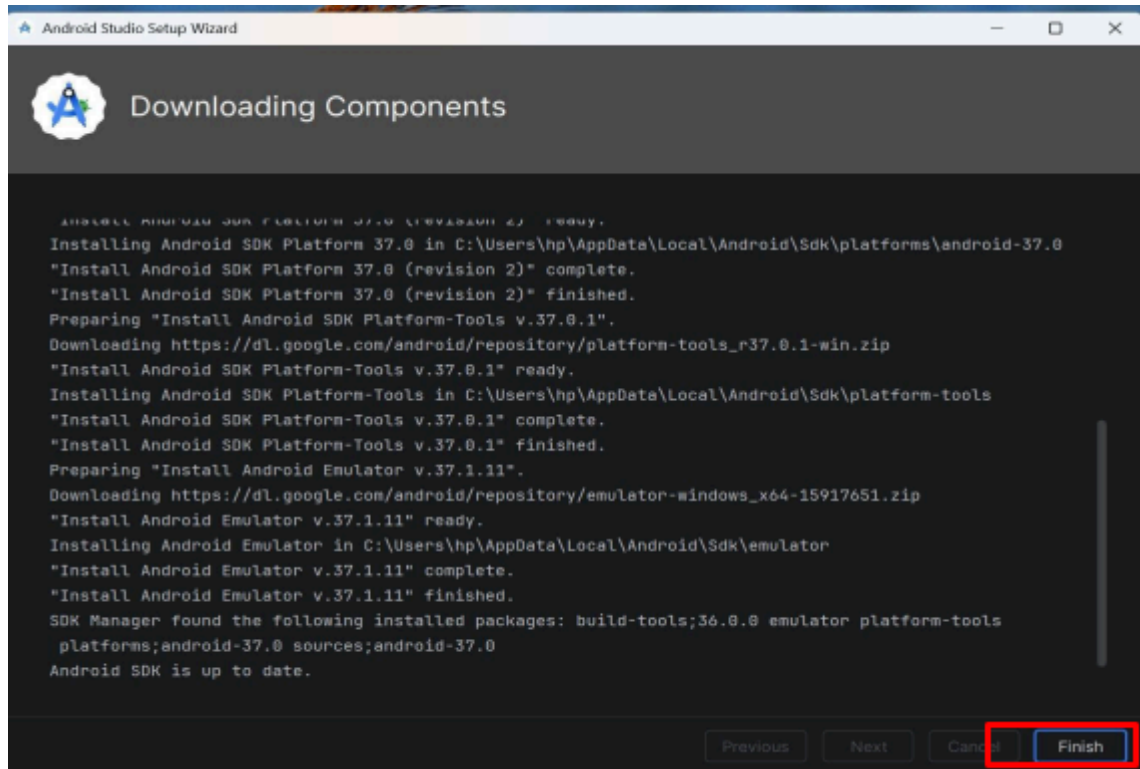
- 7.5. A Dialog Box Will Appear to confirm Download Click on **Yes**.



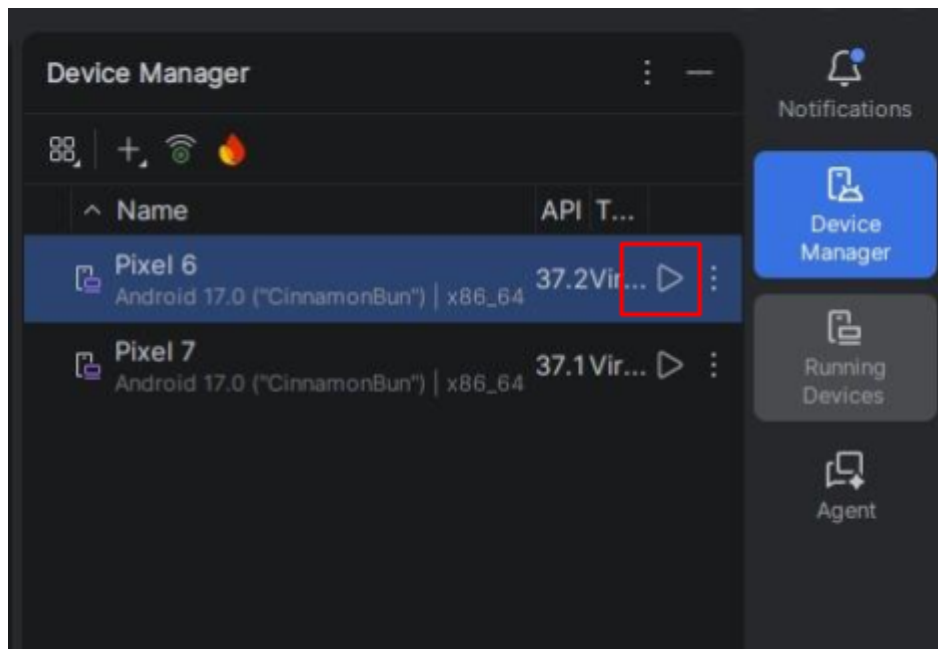
- 7.6. Downloading will start.



7.7. After Successful Download Click On **Finish**.



7.8. Now select the Pixel 6 created in above steps and then click on the ► (play) button and the emulator will start.



7.9. The emulator should look like this when started:

